

Unit-3:

$$f\left(x, y, \frac{dy}{dx}, \frac{d^2y}{dx^2}, \dots\right) = 0$$

First order linear differential equation

$$a_1(t)\dot{y} + a_0(t)y = f(t)$$

We consider the initial-value problem based on this equation, which requires a solution continuous for $0 \leq t \leq b$ satisfying

$$y(0) = y_0$$

$a_1(t), a_0(t)$ - continuous functions for $0 \leq t \leq b$.

$$\dot{y} + \frac{a_0(t)}{a_1(t)} y = \frac{f(t)}{a_1(t)}$$

$$\text{or, } \frac{dy}{dt} + q(t)y = \lambda(t)$$

$q(t), \lambda(t)$ are continuous functions. $0 \leq t \leq b$

Introduce the operator

$$Ly = \frac{dy}{dt} + q(t)y$$

We observe that L is linear, since

$$\begin{aligned} L(c_1 y_1 + c_2 y_2) &= \frac{d}{dt}(c_1 y_1 + c_2 y_2) + q(t)(c_1 y_1 + c_2 y_2) \\ &= c_1 \left(\frac{dy_1}{dt} + q(t)y_1 \right) + c_2 \left(\frac{dy_2}{dt} + q(t)y_2 \right) \\ &= c_1 Ly_1 + c_2 Ly_2 \end{aligned}$$

$$\dot{y} + q(t)y = 0 \longrightarrow \text{Homogenous equation.}$$

$$\dot{y} + q(t)y = \lambda(t) \text{ (with } \lambda(t) \neq 0)$$

$\longrightarrow$ Non-homogenous equation

★ If y_1 is a solution of the associated homogenous equation and y_2 is any solution of the non-homogenous equation

then $cy_1 + y_2$ is a solution of the non-homogeneous equation for an arbitrary constant c . *

$$L(cy_1 + y_2) = cLy_1 + Ly_2 = \lambda(t)$$

* From this family $y = cy_1 + y_2$, select one which satisfies the initial condition $y(0) = y_0$.

$$\text{i.e., } y(0) = cy_1(0) + y_2(0) = y_0$$

$$\text{or, } c = \frac{y_0 - y_2(0)}{y_1(0)}, \text{ provided } y_1(0) \neq 0$$

Existence of a solution of the initial-value problem:

$$\dot{y} + qy = 0 \text{ or, } \frac{dy}{dt} + qy = 0$$

$$\text{or, } \dot{y} = -qy$$

$$\text{or, } \frac{dy}{dt} = -qy \Rightarrow \frac{dy}{y} = -q dt$$

Integrating for $0 \leq t \leq b$,

$$\int_0^t \frac{dy}{y} = - \int_0^t q dt$$

$$\text{or, } \ln|y| - \ln|y(0)| = - \int_0^t q(\tau) d\tau$$

$$\text{or, } \ln \left| \frac{y}{y(0)} \right| = - \int_0^t q(\tau) d\tau$$

$$\text{or, } \left| \frac{y}{y(0)} \right| = \exp \left[- \int_0^t q(\tau) d\tau \right]$$

$$y_1 = \exp \left[- \int_0^t q(\tau) d\tau \right]$$

Next, we show that y_2 , a solution of the non-homogeneous equation, exists

CF (u)

complementary function

PI (V)

Particular integral

U $\rightarrow$ soln of homogeneous

Eqn $(Cy_1 + Cy_2)$

V $\rightarrow$ soln of non-homogeneous eqn

Complete soln

Y = CF + PI

f(D)y = X

f(D) = 0 $\rightarrow$ AE

AE - Auxiliary Eqn

$$\text{let } Q(t) = \exp \left[+ \int_0^t q(\tau) d\tau \right]$$

multiplying

$$\boxed{\dot{y}_2 + q y_2 = \lambda} \text{ by } Q,$$

we have,

$$Q \dot{y}_2 + q Q y_2 = \lambda Q$$

$$\Rightarrow \frac{d}{dt} (Q y_2) = \lambda Q$$

$$\Rightarrow Q y_2 = \int_0^t \lambda(\tau) Q(\tau) d\tau$$

$$\Rightarrow y_2 = \frac{1}{Q(t)} \int_0^t \lambda(\tau) Q(\tau) d\tau$$

a solution of the non-homogenous equation,

$y_1(0) = 1$ and $y_2(0) = 0$, a solution of the initial value

problem is

$$y = y_0 y_1 + y_2 = y_0 \exp \left[- \int_0^t q(\tau) d\tau \right] + \exp \left[- \int_0^t q(\tau) d\tau \right] \int_0^t \lambda(\tau) Q(\tau) d\tau$$

Ex) $x \cos x \frac{dy}{dx} + y(x \sin x + \cos x) = 1$

Soln: $\frac{dy}{dx} + q(t)y = \lambda(t)$

$$x \cos x \frac{dy}{dx} + y(x \sin x + \cos x) = 1$$

$$\frac{dy}{dx} + \frac{y(x \sin x + \cos x)}{x \cos x} = \frac{1}{x \cos x} \rightarrow (1)$$

$$IF = e^{\int \frac{x \sin x + \cos x}{x \cos x} dx}$$

$$= e^{\int \frac{\sin x}{\cos x} dx + \int \frac{1}{x} dx}$$

$$= e^{-\log |\cos x| + \log x}$$

$$= e$$

$$= e^{\log \frac{x}{\cos x}} = \frac{x}{\cos x}$$

Multiplying both sides of ① by If

$$\frac{d}{dx} \left\{ y \frac{x}{\cos x} \right\} = \frac{1}{x \cos x} \cdot \frac{x}{\cos x}$$

$$\frac{d}{dx} \left\{ \frac{xy}{\cos x} \right\} = \sec^2 x$$

Integrating,

$$\frac{xy}{\cos x} = \int \sec^2 x + C$$

$$\frac{xy}{\cos x} = \tan x + C$$

$$xy = \sin x + C(\cos x)$$

$$\text{Ex) } y^2 + \left(x - \frac{1}{y}\right) \frac{dy}{dx} = 0$$

$$\left(x - \frac{1}{y}\right) \frac{dy}{dx} = -y^2$$

$$(xy - 1) \frac{dy}{dx} = -y^3$$

$$xy - 1 = -y^3 \frac{dx}{dy}$$

$$y^3 \frac{dx}{dy} + xy = 1$$

$$\frac{dx}{dy} + \frac{1}{y^2} x = \frac{1}{y^2}$$

$$\frac{dy}{dx} + Py = Qy^n \longrightarrow \text{Bernoulli's eqn.}$$

$$\frac{1}{y^n} \frac{dy}{dx} + y^{1-n} \cdot P = Q$$

$$\text{Put } y^{1-n} = z$$

$$(1-n) y^{-n} \frac{dy}{dx} = \frac{dz}{dx}$$

$$\frac{dx}{dy} e^{\int \frac{1}{y^2} dy} + \frac{1}{y^2} x e^{\int \frac{1}{y^2} dy} = \frac{1}{y^2} e^{\int \frac{1}{y^2} dy}$$

$$\frac{dx}{dy} e^{-1/y} + \frac{1}{y^2} x e^{-1/y} = \frac{1}{y^2} e^{-1/y}$$

$$d(x e^{-1/y}) = \frac{1}{y^2} e^{-1/y} dy$$

$$\int d(x e^{-1/y}) = \int d(e^{-1/y})$$

$$x e^{-1/y} - e^{-1/y} = C$$

$$e^{-1/y} (x - 1) = C$$

$$y^{-n} \frac{dy}{dx} = \frac{1}{1-n} \frac{dz}{dx}$$

$$\frac{1}{1-n} \frac{dz}{dx} + Pz = Q$$

$$\frac{dz}{dx} + (1-n)Pz = (1-n)Q \quad \text{--- (2)}$$

$$\text{IF} = e^{\int (1-n)P dx}$$

Multiplying both sides of (2) by IF,

$$\frac{d}{dx} \left\{ z \cdot e^{\int (1-n)P dx} \right\} = (1-n)Q \cdot e^{\int (1-n)P dx}$$

$$z \cdot e^{\int (1-n)P dx} = (1-n) \int Q e^{\int (1-n)P dx} dx + C$$

$$y^{1-n} e^{\int (1-n)P dx} = (1-n) \int Q e^{\int (1-n)P dx} dx + C$$

Ex) $\frac{dy}{dx} - 5y = \sin x$ Ans: $Ce^{5x} - \frac{5}{26} \sin x - \frac{1}{26} \cos x$

$$\text{I.F} = e^{\int -5 dx} = e^{-5x}$$

$$\frac{dy}{dx} e^{-5x} - 5y e^{-5x} = \sin x e^{-5x}$$

$$\frac{d}{dx} (y e^{-5x}) = \sin x e^{-5x}$$

Integrating,

$$y e^{-5x} = \int e^{-5x} \sin x dx$$

$$y e^{-5x} = \frac{-e^{-5x} \cos x}{26} - \frac{5e^{-5x} \sin x}{26} + C$$

$$y = -\frac{1}{26} \cos x - \frac{5}{26} \sin x + C e^{5x}$$

Theorem: The necessary and sufficient condition that $Mdx + Ndy = 0$ be exact is

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

Solution: $u + \int f(y) dy = 0$
 $u = \int M dx + f(y)$ → turns out N not containing x
 $y = \text{const.}$

Necessary part: $\Rightarrow$

Suppose the equation $Mdx + Ndy = 0$ is an exact equation — (1)

Then for some function $u(x, y)$,

$Mdx + Ndy$ is a perfect differential

$$Mdx + Ndy = du \quad (2)$$

$$u = u(x, y)$$

$$du = \frac{\partial u}{\partial x} dx + \frac{\partial u}{\partial y} dy \quad (3)$$

$$M = \frac{\partial u}{\partial x}, \quad N = \frac{\partial u}{\partial y}$$

$$\frac{\partial M}{\partial y} = \frac{\partial^2 u}{\partial y \partial x}, \quad \frac{\partial N}{\partial x} = \frac{\partial^2 u}{\partial x \partial y}$$

$$\Rightarrow \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

Ex) solve $(y^2 e^{xy^2} + 4x^3) dx + (2xy e^{xy^2} - 3y^2) dy = 0$

$$\frac{\partial M}{\partial y} = \frac{\partial}{\partial y} (y^2 e^{xy^2} + 4x^3)$$

$$= 2y e^{xy^2} + 2xy^3 e^{xy^2}$$

$$\frac{\partial N}{\partial x} = \frac{\partial}{\partial x} (2xy e^{xy^2} - 3y^2)$$

$$= 2y e^{xy^2} + 2xy^3 e^{xy^2}$$

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} \Rightarrow \text{It is an exact differential}$$

$$(y^2 e^{xy^2} dx + 2xy e^{xy^2} dy) + 4x^3 dx - 3y^2 dy = 0$$

$$d(e^{xy^2}) + d(x^4) - d(y^3) = 0$$

$$\text{or } d(e^{xy^2} + x^4 - y^3) = 0$$

$$e^{xy^2} + x^4 - y^3 = C$$

After:

$$\int M dx = \int (y^2 e^{xy^2} + 4x^3) dx$$

$$= e^{xy^2} + x^4 + C$$

$$\int N dy = \int (2xy e^{xy^2} - 3y^2) dy$$

$$= e^{xy^2} - y^3 + C$$

Soln: $e^{xy^2} + x^4 - y^3 = C$

Repeated terms should be written only once.

Ex) $\tan y dx + \tan x dy = 0$

$$\left. \begin{aligned} \frac{\partial M}{\partial y} &= \frac{\partial}{\partial y} (\tan y) = \sec^2 y \\ \frac{\partial N}{\partial x} &= \frac{\partial}{\partial x} (\tan x) = \sec^2 x \end{aligned} \right\} \frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$$

$$\tan y dx + \tan x dy = 0$$

$$\underbrace{\sin y \cos x dx}_M + \underbrace{\sin x \cos y dy}_N = 0$$

$$\left. \begin{aligned} \frac{\partial M}{\partial y} &= \frac{\partial}{\partial y} (\sin y \cos x) = \cos y \cos x \\ \frac{\partial N}{\partial x} &= \frac{\partial}{\partial x} (\sin x \cos y) = \cos x \cos y \end{aligned} \right\} \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

$$d(\sin x \sin y) = 0$$

$$\sin x \sin y = C$$

Note: $M(x, y) dx + N(x, y) dy = 0$ — let this not be exact.

$\mu(x, y) \{ M(x, y) dx + N(x, y) dy \} \equiv$ an exact differential.
 $= du$

Integrating factor (I.F)

Ex) Solve $x dx + y dy + \frac{x dy - y dx}{x^2 + y^2} = 0$

$$x dx + y dy = \frac{1}{2} d(x^2 + y^2)$$

$$\frac{x dy - y dx}{x^2} = d\left(\frac{y}{x}\right)$$

$$\frac{x dy - y dx}{x^2 + y^2} = \frac{\frac{x dy - y dx}{x^2}}{\frac{x^2 + y^2}{x^2}} = \frac{d\left(\frac{y}{x}\right)}{1 + \left(\frac{y}{x}\right)^2}$$

$$= d\left(\tan^{-1}\left(\frac{y}{x}\right)\right)$$

$$\frac{1}{2} d(x^2 + y^2) + d\left(\tan^{-1}\left(\frac{y}{x}\right)\right) = 0$$

$$\frac{1}{2} (x^2 + y^2) + \tan^{-1}\left(\frac{y}{x}\right) = c'$$

$$x^2 + y^2 + 2 \tan^{-1}\left(\frac{y}{x}\right) = c$$

Rules for determining I.F.:

Rule-1) If $M dx + N dy = 0$ is homogenous but not exact and if $Mx + Ny \neq 0$, then $\frac{1}{Mx + Ny}$ is an I.F.

Rule-2) If $\frac{1}{N} \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right)$ is a function of x alone, $f(x)$, say, then $e^{\int f(x) dx}$ is an I.F. of $M dx + N dy = 0$.

$$* \frac{1}{N} \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) = f(x)$$

Ex) $(x^3 + xy^4) dx + 2y^3 dy = 0$. Ans: $e^{x^2} (x^2 - 1 + y^4) = c$

$$\left. \begin{aligned} \frac{\partial M}{\partial y} &= \frac{\partial}{\partial y} (x^3 + xy^4) = 4xy^3 \\ \frac{\partial N}{\partial x} &= \frac{\partial}{\partial x} (2y^3) = 0 \end{aligned} \right\} \frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x} \Rightarrow \text{Not exact}$$

$$\frac{1}{N} \left(\left(\frac{\partial M}{\partial y} \right) - \left(\frac{\partial N}{\partial x} \right) \right) = \frac{1}{2y^3} \cdot 4xy^3 = 2x \rightarrow f(x)$$

$$I.F = e^{\int 2x dx} = e^{x^2}$$

Multiply by the I.F e^{x^2}

$$(x^3 + xy^4) dx \times e^{x^2} + 2y^3 e^{x^2} dy = 0$$

$$x^3 e^{x^2} dx + xy^4 e^{x^2} dx + 2y^3 e^{x^2} dy = 0$$

$$\int M dx = \int (x^3 + xy^4) e^{x^2} dx$$

$$\begin{aligned} & \int x^3 \cdot x \cdot e^{x^2} dx + \frac{1}{2} \int y^4 e^{x^2} dx \\ &= \frac{1}{2} \int t \cdot e^t dt + \frac{1}{2} \int y^4 e^{x^2} dx \\ &= \frac{1}{2} (t e^t - \int e^t dt) + \frac{1}{2} \int y^4 e^{x^2} dx \\ &= \frac{x^2 e^{x^2}}{2} - \frac{e^{x^2}}{2} + \frac{1}{2} \int y^4 e^{x^2} dx \\ &= \frac{e^{x^2}}{2} [x^2 + y^4 - 1] \end{aligned}$$

Rule-3) If $\frac{1}{M} \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right)$ is a function of y alone, say $\phi(y)$

then $e^{-\int \phi(y) dy}$ is an IF of $M dx + N dy = 0$

Ex) $(2xy^4 e^y + 2xy^3 + y) dx + (x^2 y e^y - x^2 y^2 - 3x) dy = 0$

$$\frac{\partial M}{\partial y} = 8xy^3 e^y + 2xy^4 e^y + 6xy^2 + 1$$

$$\frac{\partial N}{\partial x} = 2xy^2 e^y - 2xy^2 - 3$$

Ans: $x^2 e^y + \frac{x^2}{y} + \frac{x}{y^3} = C$

$$\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} =$$

~~Wrong Question~~
Unsolvable.
Correct one later.

Proof of Rule - (ii)

$$\frac{1}{N} \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) = f(x)$$

$$\text{or } \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} + Nf(x)$$

$$\Rightarrow \frac{\partial M}{\partial y} e^{\int f(x) dx} = \frac{\partial N}{\partial x} e^{\int f(x) dx} + Nf(x) e^{\int f(x) dx}$$

$$\frac{\partial}{\partial y} (M e^{\int f(x) dx}) = \frac{\partial}{\partial x} (N e^{\int f(x) dx})$$

$\Rightarrow$ The equation $\{ M e^{\int f(x) dx} \} dx + \{ N e^{\int f(x) dx} \} dy = 0$ is an exact equation.

Proof of Rule - (iii)

$$\frac{1}{M} \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) = \phi(y)$$

$$\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} = M\phi(y)$$

$$\frac{\partial M}{\partial y} - M\phi(y) = \frac{\partial N}{\partial x}$$

$$e^{-\int \phi(y) dy} \frac{\partial M}{\partial y} - M\phi(y) e^{-\int \phi(y) dy} = \frac{\partial N}{\partial x} e^{-\int \phi(y) dy}$$

$$\text{or } \frac{\partial}{\partial y} \{ M e^{-\int \phi(y) dy} \} = \frac{\partial}{\partial x} \{ N e^{-\int \phi(y) dy} \}$$

$$\Rightarrow M_1 dx + N_1 dy = 0$$

$$\Rightarrow (M dx + N dy) e^{-\int \phi(y) dy} = 0$$

Ex) Solve $(2xy^4 e^y + 2xy^3 + y) dx + (x^2 y^4 e^y - x^2 y^2 - 3x) dy = 0$

$$M dx + N dy = 0$$

$$\frac{\partial M}{\partial y} = 8xy^3 e^y + 2xy^4 e^y + 6xy^2 + 1$$

$$\frac{\partial N}{\partial x} = 2xy^4 e^y - 2xy^2 - 3$$

NUMERICAL METHODS

$$\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} = 8xy^3e^y + 8xy^2 + 4$$

$$\frac{1}{M} \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) = \frac{8xy^3e^y + 8xy^2 + 4}{2xy^4e^y + 2xy^3 + y} = \frac{4y(2xy^3e^y + 2xy^2 + 1)}{y(2xy^3e^y + 2xy^2 + 1)} = 4/y$$

An I.F is $e^{-\int \phi(y) dy} = e^{-\int \frac{4}{y} dy}$

$$= e^{-4 \ln y} = 1/y^4$$

Multiplying by I.F,

$$\left(2xe^y + \frac{2x}{y} + \frac{1}{y^3} \right) dx + \left(x^2e^y - \frac{x^2}{y^2} - \frac{3x}{y^4} \right) dy = 0$$

$$d \left(x^2e^y + \frac{x^2}{y} + \frac{x}{y^3} \right) = 0$$

$$x^2e^y + \frac{x^2}{y} + \frac{x}{y^3} = C$$

Rule-IV If $Mx - Ny \neq 0$, then $\frac{1}{Mx - Ny}$ is an IF of the equation $Mdx + Ndy = 0$, where M and N can be written in the form $M = yf_1(xy)$, $N = xf_2(xy)$

$$Mx - Ny = xy(f_1(xy) - f_2(xy)) \neq 0$$

$$Mdx + Ndy = \frac{1}{2} \left\{ (Mx + Ny) \left(\frac{dx}{x} + \frac{dy}{y} \right) + (Mx - Ny) \left(\frac{dx}{x} - \frac{dy}{y} \right) \right\}$$

$$\frac{Mdx + Ndy}{Mx - Ny} = \frac{1}{2} \left\{ \frac{Mx + Ny}{Mx - Ny} d(\log xy) + d\left(\log \frac{x}{y}\right) \right\} \equiv$$

$$= \frac{1}{2} \left\{ \frac{f_1 + f_2}{f_1 - f_2} d(\log xy) + d\left(\log \left(\frac{x}{y}\right)\right) \right\}$$

$$= \frac{1}{2} \left\{ F(xy) d(\log xy) + d\left(\log \frac{x}{y}\right) \right\}$$

$$= \frac{1}{2} \left\{ F \{ e^{\log xy} \} d(\log(xy)) + d\left(\log \frac{x}{y}\right) \right\}$$

Ex) Solve $(x^2y^2 + xy + 1)ydx - (x^2y^2 - xy + 1)x dy = 0$

$$\frac{Mdx + Ndy}{Mdx - Ny} = \frac{1}{2} \left\{ \frac{Mx + Ny}{Ma - Ny} d \log xy + d \log \frac{x}{y} \right\}$$

$$= \frac{1}{2} \left\{ \frac{x^2y^2 + xy + 1}{x^2y^2 + 1} \right\}$$

$$\frac{(x^2y^2 + xy + 1)ydx - (x^2y^2 - xy + 1)x dy}{xy(2x^2y^2 + 2)}$$

$$= \frac{1}{2} \left\{ \frac{xy(2xy)}{xy(2x^2y^2 + 2)} d \log xy + d \log \frac{x}{y} \right\} \frac{(x^2y^2 + y + \frac{1}{x})dx}{-(x^2y - x + \frac{1}{y})dy}$$

$$= \frac{1}{2} \left\{ \frac{xy}{x^2y^2 + 1} \cdot \frac{1}{xy} d(xy) + d \left(\log \frac{x}{y} \right) \right\}$$

$$= \frac{1}{2} \left\{ \frac{d(xy)}{x^2y^2 + 1} + d \log \left(\frac{x}{y} \right) \right\}$$

$$\Rightarrow \frac{1}{2} \left\{ \tan^{-1}(xy) + \log \frac{x}{y} \right\} = c'$$

$$\tan^{-1}(xy) + \log \frac{x}{y} = c$$

Rule-V The equation $x^\alpha y^\beta (mydx + nx dy) + x^{\alpha_1} y^{\beta_1} (m_1 y dx + n_1 x dy) = 0$

(where $\alpha, \beta, \alpha_1, \beta_1, m, n, m_1, n_1$ are constants and $mn_1 - m_1n \neq 0$), has an I.F of the form $x^h y^k$, where h and k are constants given by

$$\frac{h+1+\alpha}{m} = \frac{k+1+\beta}{n}$$

$$\frac{h+1+\alpha_1}{m_1} = \frac{k+1+\beta_1}{n_1}$$

Ex) Solve $3ydx - 2x dy + x^2 y^{-1} (10ydx - 6x dy) = 0$

Soln If $x^h y^k$ is an I.F, then

$$x^h y^k (3ydx - 2x dy) + x^h y^k x^2 y^{-1} (10ydx - 6x dy) = 0$$

must be exact and hence

$$\frac{\partial}{\partial y} \{ 3y^{k+1}x^h + 10x^{h+2}y^k \} = \frac{\partial}{\partial x} \{ -2x^{h+1}y^k - 6x^{h+3}y^{k-1} \}$$

$$3(k+1)y^k x^h + 10k x^{h+2} y^{k-1} = -2(h+1)x^h y^k - 6(h+3)x^{h+2} y^{k-1}$$

$$3(k+1) + 2(h+1) = 0 \quad 3k + 2h + 5 = 0 \Rightarrow 9k + 6h + 15 = 0$$

$$10k + 6(h+3) = 0 \quad 10k + 6h + 18 = 0$$

$$\boxed{k = -3; h = 2}$$

$$I.F = \frac{x^2}{y^3}$$

Multiplying by IF,

$$3\frac{x^2}{y^4} dx - \frac{2x^3}{y^3} dy + \frac{x^4}{y^4} (10y dx - 6x dy) = 0$$

$$d\left(\frac{x^3}{y^2} + \frac{2x^5}{y^3}\right) = 0$$

$$\frac{x^3}{y^2} + \frac{2x^5}{y^3} = C$$

Two families of curves with the property that each member of either family cuts each member of the other family at right angles are said to be orthogonal trajectories of each other.

Ex) Find the orthogonal trajectories of the ~~curves~~^{series} of hyperbolas $xy = a^2$.

$$x \frac{dy}{dx} + y = 0$$

The differential equation corresponding to the orthogonal trajectories of curve ① is obtained by replacing $\left(\frac{dy}{dx}\right)$ by $\left(-\frac{dx}{dy}\right)$

$$x\left(-\frac{dx}{dy}\right) + y = 0$$

$$x dx - y dy = 0$$

$$x^2 - y^2 = b^2$$

Ex) Find the orthogonal trajectories of the family of hyperboloids

$$x^{2/3} + y^{2/3} = a^{2/3}, \text{ where } a \text{ is a parameter.}$$

Soln.

$$\frac{2}{3} x^{-1/3} + \frac{2}{3} y^{-1/3} \frac{dy}{dx} = 0$$

$$+ \frac{dx}{dy} = \left(\frac{x}{y} \right)^{-1/3} = \left(\frac{y}{x} \right)^{1/3}$$

$$x^{1/3} dx - y^{1/3} dy = 0$$

$$\frac{3}{4} (x^{4/3} - y^{4/3}) = C^{1/3}$$

$$y^{4/3} - x^{4/3} = C^{1/3}$$

Ex) $\frac{dy}{dx} - y = 2y^3$

$$dy - y dx - 2y^3 dx = 0$$

$$dy + (-y - 2y^3) dx = 0$$

$$M = -y - 2y^3$$

$$N = 1$$

$$\frac{\partial M}{\partial y} = -1 - 6y^2$$

$$\frac{\partial N}{\partial x} = 0$$

$$\frac{1}{M} \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) = \frac{-1 - 6y^2}{-y - 2y^3}$$

$$\frac{1}{M} \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) = \frac{-1 - 6y^2}{-y - 2y^3} = \frac{1 + 6y^2}{y(1 + 2y^2)}$$

$$\frac{dy}{dx} = y(2y^2 + 1)$$

$$\frac{dy}{y(2y^2 + 1)} = dx$$

$$\frac{dy}{dx} \cdot \frac{1}{y^3} = 2 + \frac{1}{y^2}$$

$$\frac{1}{2} \frac{dz}{dx} = 2 - z$$

$$\frac{1}{2} \frac{dz}{dx} e^{2x} = (2 - z) e^{2x}$$

$$-\ln(2 - z) = 2x + C$$

$$-\frac{dx}{dy} - y = 2y^3$$

$$-\frac{dx}{dy} = 2y(2y^2 + 1)$$

$$-dx = \frac{dy}{dy} (4y^3 + y)$$

$$-x = \frac{1}{2} y^4 + \frac{y^2}{2} + C$$

$$y^4 + y^2 + 2x = C \quad \text{D.T.}$$

$$z = -1/y^2$$

$$\frac{dz}{dx} = \frac{2}{y^3} \frac{dy}{dx}$$

$$\frac{dz}{dx} + 2z = 4$$

$$e^{2x} \frac{dz}{dx} + 2z e^{2x} = 4e^{2x}$$

$$z e^{2x} = 2e^{2x} + C$$

$$e^{2x} \left(2 + \frac{1}{y^2} \right) = C$$

Existence and Uniqueness:

First order differential equation $F(t, y, \dot{y}) = 0$

$$\Rightarrow \frac{dy}{dt} = f(t, y)$$

A corresponding initial value problem is to find a function $y(t)$ which is continuous and has a continuous derivative in $0 \leq t \leq a$ which satisfies $\dot{y} = f(t, y)$ and $y(0) = y_0$.

Definition: A function $f(t, y)$ satisfies a Lipschitz condition

in $R = \{ (t, y) \mid 0 \leq t \leq b, |y - y_0| \leq c \}$ if there exists a constant $k > 0$ such that

$$|f(t, y_1) - f(t, y_2)| \leq k|y_1 - y_2|$$

for each (t, y_1) and (t, y_2) in R .

$$|y_1 + y_2| \leq |2c + 2|y_0|$$

Ex) The function $f(t, y) = ty^2$ satisfies a Lipschitz condition in

$$0 \leq t \leq b < \infty, |y - y_0| \leq c < \infty$$

$$|ty_1^2 - ty_2^2| = |t| |y_1 + y_2| |y_1 - y_2| \leq b(2c + 2|y_0|) |y_1 - y_2|$$

because,

$$\begin{aligned} |y_1 + y_2| &= |y_1 - y_0 + y_2 - y_0 + 2y_0| \\ &\leq |y_1 - y_0| + |y_2 - y_0| + 2|y_0| \\ &\leq 2c + 2|y_0| \end{aligned}$$

Ex) The function $f(t, y) = \sqrt{y}$ does not satisfy a Lipschitz condition for $0 \leq t \leq b, |y - t| \leq 1$

because,

$$\begin{aligned} |\sqrt{y_1} - \sqrt{y_2}| &= \frac{|\sqrt{y_1} - \sqrt{y_2}| |\sqrt{y_1} + \sqrt{y_2}|}{\sqrt{y_1} + \sqrt{y_2}} \\ &= \frac{|y_1 - y_2|}{\sqrt{y_1} + \sqrt{y_2}} \end{aligned}$$

$\sqrt{y_1} + \sqrt{y_2}$ is not bounded in the given region,

Theorem: If $f(t, y)$ has a continuous partial derivative f_y in $R = \{(t, y) | 0 \leq t \leq b < \infty, |y - y_0| \leq c < \infty\}$, then it satisfies a Lipschitz condition in R .

Theorem: Let $f(t, y)$ be ~~conditio~~ continuous and satisfy a Lipschitz condition in $R = \{(t, y) | 0 \leq t \leq b, |y - y_0| \leq c\}$.

Let $M = \max |f(t, y)|$ in R . Then the initial value problem

$\dot{y} = f(t, y), y(0) = y_0$ has a unique solution for

$$0 \leq t \leq \min \left[b, \frac{c}{M} \right]$$

~~Ex)~~ D.T

$$x = a(1 - \cos \theta)$$

$$\frac{dx}{d\theta} = a \sin \theta$$

Eliminating a , we have,

$$\frac{\frac{dx}{d\theta}}{x} = \frac{\sin \theta}{1 - \cos \theta}$$

$$\frac{1}{x} \frac{dx}{d\theta} = \frac{2 \sin \theta/2 \cos \theta/2}{2 \sin^2 \theta/2}$$

$$\boxed{\frac{1}{x} \frac{dx}{d\theta} = \cot \theta/2}$$

Substituting $dx/d\theta$ by $-x^2 d\theta/dx$

$$\frac{1}{x} \left(-x^2 \frac{d\theta}{dx} \right) = \cot \theta/2$$

$$-x \frac{d\theta}{dx} = \cot \theta/2$$

$$\frac{dx}{x} = 2 \left(\frac{-1/2 \sin \theta/2}{\cos \theta/2} \right) d\theta$$

$$\log x = 2 \log \cos \theta/2 + \log c$$

$$\log x = \log (c (2 \cos^2 \theta/2))$$

$$x = c(1 - \cos \theta)$$

$$x = r \cos \theta$$

$$y = r \sin \theta$$

$$dx = -r \sin \theta d\theta + \cos \theta dr$$

$$dy = r \cos \theta d\theta + \sin \theta dr$$

$$\frac{dy}{dx} = \frac{r \frac{dy}{dr} + \sin \theta}{-r \frac{dx}{dr} + \cos \theta}$$

$$\frac{dy}{dx} = \frac{r \frac{dy}{dr} + \sin \theta}{-r \frac{dx}{dr} + \cos \theta}$$

$$\Rightarrow -x \frac{dy}{dx}$$

$$= -x \frac{dy}{dx}$$

$$-x^2 \frac{d\theta}{dx}$$

Theorem-1: If ~~f(t,y)~~ $f(t,y)$

$$f_y \text{ in } R = \{ (t,y) \mid 0 \leq t \leq b < \infty, |y-y_0| \leq c \}$$

then f satisfies Lipschitz condition in R .

Proof: $f(t,y_1) - f(t,y_2) = f_y(t,\eta)(y_1 - y_2)$

where η lies between y_1 and y_2 .

Therefore (t,η) is in R and $f_y(t,\eta)$ is bounded in R

Hence, $|f(t,y_1) - f(t,y_2)| \leq K|y_1 - y_2|$

where, $K = \max |f_y|$ in R .

Theorem-2: $f(t,y)$ continuous satisfies Lipschitz condition

in R . Let $M = \max |f(t,y)|$ in R . Then the initial value

problem (IVP) has unique solution for

$$0 \leq t \leq \min \left[b, \frac{c}{M} \right]$$

Ex $\frac{dy}{dx} = y + e^{2x}$, $y(0) = 2$

$f(x,y) = y + e^{2x}$

is continuous w.r.t x and y .

$$|f(x,y_1) - f(x,y_2)| = |y_1 - y_2|$$

By the uniqueness theorem, the IVP has a unique solution

$y(x) = e^{2x} + 1$ is the only solution.

Ex Consider a non-linear IVP $\frac{dy}{dx} = x \sin y$

$$D = \{ (x,y) : |x| \leq 2, -\infty < y < \infty \}$$

$$\frac{dy}{dx} = x \sin y; y(0) = 1$$

Soln: $f(x,y) = x \sin y$ is continuous on D .

$\max(x,y) < \infty$

$$\left| \frac{\partial f}{\partial y} \right| = \max_{(x,y) \in D} |x \cos y| = 2$$

$\Rightarrow x \sin y$ is Lipschitz continuous on D w.r.t y with Lipschitz constant $L = 2$.
 f satisfies conditions of uniqueness theorem.

Ex) Consider the IVP (linear) $\frac{dy}{dx} = \frac{2}{x} xy = \frac{2}{x} y, y(0) = 0$

$$y = cx^2$$

$$\frac{2}{x} (y_1 - y_2) =$$

Picard's Theorem:

Let $f(x, y)$ and $\frac{\partial f}{\partial y}$ be continuous at all points in a closed rectangular region R . If (x_0, y_0) is any point in the interior of R , then $\exists$ a positive number h such that the IVP

$$y' = f(x, y), y(x_0) = y_0$$

has a unique solution for all x in the interval $[x_0 - h, x_0 + h]$

Ex) Test for uniqueness of solutions of the IVP

$$(x - 2y + 1) dy - (3x - 6y + 2) dx = 0, y(0) = 0$$

over the rectangle $R: |x| \leq \frac{1}{4}, |y| \leq \frac{1}{4}$

Determine the interval for x , over which solution is guaranteed

Soln: $f(x, y) = \frac{3x - 6y + 2}{x - 2y + 1}, x_0 = 0, y_0 = 0$

$$|f(x, y)| = \frac{|3x - 6y + 2|}{|x - 2y + 1|} < \frac{3|x| + 6|y| + 2}{1 - |x| - 2|y|}$$

$$< \frac{3(\frac{1}{4}) + 6(\frac{1}{4}) + 2}{1 - \frac{1}{4} - 2(\frac{1}{4})} = \frac{8 + \frac{3}{4}}{\frac{1}{4}} = 32 + 3 = 35 = M$$

$$f_y = \left| \frac{(x-2y+1)(-6) - (-2)(3x-6y+2)}{(x-2y+1)^2} \right| = \left| \frac{-2}{(x-2y+1)^2} \right|$$

$$< \frac{2}{\left(\frac{1}{4}\right)^2} = 32 = M$$

~~Thus,~~

Also, $h = \min \left\{ \frac{1}{4}, \frac{1}{68} \right\} = \frac{1}{68}$

Therefore the IVP has a unique solution atleast for all x in the interval, $|x| \leq \frac{1}{68}$

$$-\frac{1}{68} \leq x \leq \frac{1}{68}$$

Homogenous Linear Differential Equations with constant coefficients:

$$a_0 \frac{d^n y}{dx^n} + a_1 \frac{d^{n-1} y}{dx^{n-1}} + a_2 \frac{d^{n-2} y}{dx^{n-2}} + \dots + a_{n-1} \frac{dy}{dx} + a_n y = 0 \rightarrow \textcircled{1}$$

Assuming that $y = e^{mx}$ is a solution for certain m ,

$$\frac{dy}{dx} = m e^{mx}; \quad \frac{d^2 y}{dx^2} = m^2 e^{mx}; \quad \dots \quad \frac{d^n y}{dx^n} = m^n e^{mx}$$

Substituting in $\textcircled{1}$, we obtain

$$a_0 m^n e^{mx} + a_1 m^{n-1} e^{mx} + \dots + a_{n-1} m e^{mx} + a_n e^{mx} = 0$$

$$e^{mx} (a_0 m^n + a_1 m^{n-1} + \dots + a_{n-1} m + a_n) = 0$$

Since, $e^m \neq 0$, we obtain the auxiliary equation.

$$a_0 m^n + a_1 m^{n-1} + \dots + a_{n-1} m + a_n = 0 \rightarrow \textcircled{2}$$

Case-1) Distinct Real roots.

$$m_1, m_2, \dots, m_n$$

$$e^{m_1 x}, e^{m_2 x}, \dots, e^{m_n x}$$

General solution is

$$y = C_1 e^{m_1 x} + C_2 e^{m_2 x} + \dots + C_n e^{m_n x}$$

$C_1, C_2, C_3, \dots, C_n$ are arbitrary constants.

Ex) $\frac{d^2y}{dx^2} - 3\frac{dy}{dx} + 2y = 0$,

$$m^2 - 3m + 2 = 0 \Rightarrow (m-1)(m-2) = 0$$

$$\Rightarrow m = 1, 2$$

$$y = C_1 e^x + C_2 e^{2x}$$

Ex). $\frac{d^3y}{dx^3} + 4\frac{d^2y}{dx^2} + \frac{dy}{dx} + 6y = 0$

The auxiliary equation is $m^3 - 4m^2 + m + 6 = 0$

$$(m+1)(m^2 - 5m + 6) = 0$$

$$(m+1)(m-2)(m-3) = 0$$

$$m = -1, 2, 3$$

General solution $\therefore y = C_1 e^{-x} + C_2 e^{2x} + C_3 e^{3x}$

Case-2: Repeated real roots.

Ex) Consider the differential equation

$$\frac{d^2y}{dx^2} - 6\frac{dy}{dx} + 9y = 0 \rightarrow \textcircled{1}$$

Auxiliary equation is $m^2 - 6m + 9 = 0$

$$(m-3)^2 = 0$$

$$\text{or, } m = 3, 3$$

Let $y = e^{3x} v$, where v is to be determined.

$$\frac{dy}{dx} = 3e^{3x} v + e^{3x} \frac{dv}{dx}$$

$$\begin{aligned} \frac{d^2y}{dx^2} &= 9e^{3x} v + 3e^{3x} \frac{dv}{dx} + 3e^{3x} \frac{dv}{dx} + e^{3x} \frac{d^2v}{dx^2} \\ &= e^{3x} \frac{d^2v}{dx^2} + 6e^{3x} \frac{dv}{dx} + 9e^{3x} v \end{aligned}$$

Substituting into eqn $\textcircled{1}$, we get,

$$e^{3x} \frac{dv}{dx} + 6e^{3x} \frac{dv}{dx} + 9e^{3x} v - 6e^{3x} \frac{dv}{dx} - 18e^{3x} v + 9e^{3x} v = 0$$

$$e^{3x} \frac{d^2v}{dx^2} = 0$$

Let $w = \frac{dv}{dx}$, we get the first order equation

$$e^{3x} \frac{dw}{dx} = 0 \Rightarrow \frac{dw}{dx} = 0, \text{ or } w = c.$$

$$\frac{dv}{dx} = c$$

$$dv = c dx$$

$$v = cx + k.$$

Taking $c=1$, $v = x + C_0$

$$y = e^{3x} (x + C_0)$$

Fundamental solution: $y = x e^{3x}$

Theorem: Then n^{th} order homogenous linear differential equation (1) with constant coefficients. If the auxiliary eqn (A.E) (2) has the real roots α occurring k times, then the part of general solⁿ corresponding to this k -fold repeated roots is

$$(C_1 + C_2 x + C_3 x^2 + \dots + C_k x^{k-1}) e^{\alpha x}.$$

Tutorial:

A. Test whether the equation is exact or not and hence solve it.

$$1. (1 + 2xy \cos x^2 - 2xy) dx + (\sin x^2 - x^2) dy = 0$$

$$2. [y(1 + \frac{1}{x}) + \cos y] dx + [x + \log x - x \sin y] dy = 0$$

B. Find the integrating factor and solve the following differential equation.

$$1. x^2 \frac{dy}{dx} + xy = \sqrt{1 - x^2 y^2}$$

$$2. (y + 2xy^2) dx + (x - x^2 y) dy = 0$$

C. Solve the Bernoulli's equation

$$1. \frac{dy}{dx} + x \sin 2y = x^3 \cos^2 y$$

$$2. \frac{dy}{dx} + \frac{y}{x} = \frac{y^2}{x^2}$$

D. Find the orthogonal trajectories of $r^n = a^n \cos(n\theta)$

Solutions:

$$A) i) \frac{\partial M}{\partial y} = \frac{\partial}{\partial y} (1 + 2xy \cos x^2 - 2xy) = 2x \cos x^2 - 2x$$

$$\frac{\partial N}{\partial x} = \frac{\partial}{\partial x} (\sin x^2 - x^2) = 2x \cos x^2 - 2x$$

Exact

$$I.F. = e^{\int \frac{1}{\sin x^2 - x^2} dx}$$

$$(1 + 2xy \cos x^2 - 2xy) e^{\int \frac{1}{\sin x^2 - x^2} dx} + (\sin x^2 - x^2) e^{\int \frac{1}{\sin x^2 - x^2} dx} dy = 0$$

$$y e^{\int \frac{1}{\sin x^2 - x^2} dx} (2x \cos x^2 - 2x + (\sin x^2 - x^2) \frac{1}{\sin x^2 - x^2})$$

$$y (d(\sin x^2 - x^2)) e^{\int \frac{1}{\sin x^2 - x^2} dx} = 0$$

$$(1 + 2xy \cos x^2 - 2xy) dx + (\sin x^2 - x^2) dy = 0$$

$$dx + d(y(\sin x^2 - x^2)) = 0$$

$$\boxed{x + y(\sin x^2 - x^2) = C}$$

$$A. 2) \frac{\partial M}{\partial y} = \frac{\partial}{\partial y} \left(y \left(1 + \frac{1}{x} \right) + \cos y \right) = 1 + \frac{1}{x} - \sin y$$

Exact

$$\frac{\partial N}{\partial x} = \frac{\partial}{\partial x} (x + \log x - x \sin y) = 1 + \frac{1}{x} - \sin y$$

$$\boxed{xy + y \log x + x \cos y = C}$$

$$B) 1) x^2 \frac{dy}{dx} + xy = \sqrt{1-x^2y^2}$$

$$x(xdy + ydx) = \sqrt{1-x^2y^2} dx$$

$$x d(xy) = \sqrt{1-x^2y^2} dx$$

$$\frac{d(xy)}{\sqrt{1-(xy)^2}} = \frac{dx}{x}$$

$$\boxed{\sin^{-1}(xy) - \log x = C}$$

Exact

No IF required.

$$2) (y + xy^2) dx + (x - x^2y) dy = 0$$

$$\frac{\partial M}{\partial y} = 1 + 2xy \quad \text{IF} = (Mx - Ny)$$

$$\frac{\partial N}{\partial x} = 1 - 2xy$$

$$\text{I.F} = \frac{1}{xy + x^2y^2 - xy + x^2y^2} = \frac{1}{2x^2y^2}$$

$$\frac{y + xy^2}{2x^2y^2} dx + \frac{x - x^2y}{2x^2y^2} dy = 0$$

$$\frac{1}{2x^2y} dx + \frac{dx}{2x} + \frac{dy}{2xy^2} - \frac{dy}{2y} = 0$$

$$\int \frac{(y dx + x dy)}{2x^2y^2} + \frac{1}{2} (\log x) - \frac{1}{2} \log y = C$$

$$\int \frac{d(xy)}{(xy)^2} + \log x - \log y = C$$

$$\boxed{-\frac{1}{xy} + \log x - \log y = C}$$

$$C.1) \frac{dy}{dx} + x \sin 2y = x^3 \cos y \Rightarrow \frac{dy}{dx} = x \cos y (x^2 \cos y - 2 \sin y)$$

$$\text{let } x^3 = z$$

$$3x^2 dx = dz$$

$$\frac{dy}{dz} \times 3z^{2/3} + z^{1/3} \sin 2y = z \cos y$$

$$\frac{dy}{dx} \sec y + x(2 \tan y) = x^3$$

$$d(\cancel{x^2 \tan y}) \quad \tan y = z \Rightarrow \sec y dy = dz$$

$$\frac{dz}{dx} + 2xz = x^3 \quad \text{IF} = e^{\int 2x dx} = e^{x^2}$$

$$\frac{dz}{dx} e^{x^2} + 2xz e^{x^2} = x^3 e^{x^2}$$

$$\int d(e^{x^2} \times z) = \int x^3 e^{x^2} dx \quad \begin{matrix} x^2 = t \\ 2x dx = dt \end{matrix}$$

$$e^{x^2} \times \tan y = \frac{1}{2} \int t e^t dt = \frac{1}{2} [t e^t - e^t] + C$$

$$\tan y = \frac{1}{2} e^{-x^2} [x^2 - 1] + C e^{-x^2}$$

$$\boxed{\tan y = \frac{1}{2} (x^2 - 1) + C e^{-x^2}}$$

$$2) \frac{dy}{dx} + \frac{y}{x} = \frac{y^2}{x^2}$$

$$\frac{1}{y} = z \Rightarrow \frac{-1}{y^2} dy = dz \Rightarrow dy = \frac{dz}{-z^2}$$

$$-\frac{1}{z^2} \frac{dz}{dx} + \frac{1}{xz} = \frac{1}{x^2 z^2}$$

$$\frac{dz}{dx} - \frac{z}{x} + \frac{1}{x^2} = 0$$

$$\frac{dz}{dx} - \frac{z}{x} = -\frac{1}{x^2}$$

$$\text{IF} = e^{\int -\frac{1}{x} dx} = e^{-\ln x} = \frac{1}{x}$$

$$\frac{1}{x} \frac{dz}{dx} - \frac{z}{x^2} = -\frac{1}{x^3}$$

$$\frac{1}{4} x^{-1} \quad -\frac{1}{x} \quad -\frac{2}{x^3}$$

$$\int d\left(\frac{1}{x} z\right) = \frac{1}{2x^2} + C$$

$$\frac{1}{xy} = \frac{1}{2x^2} + C$$

$$\boxed{\frac{1}{y} = \frac{1}{2x} + Cx}$$

$$D. x^n = a^n \cos(n\theta)$$

$$x^{n-1} dx = a^n - \sin(n\theta) (n) d\theta$$

$$\frac{dx}{d\theta} = \frac{-a^n \sin(n\theta)}{x^{n-1}} = \frac{-\sin(n\theta)}{x^{n-1}} \frac{x^n}{\cos n\theta}$$

$$\frac{dx}{d\theta} = -x \tan(n\theta)$$

$$\Rightarrow \frac{x^{\frac{1}{n}}}{x^{\frac{1}{n}}} \frac{d\theta}{dx} = +x \tan(n\theta)$$

$$\frac{d\theta}{dx} = x^{-\frac{1}{n}} \tan(n\theta)$$

$$\frac{d\theta}{\tan(n\theta)} = x^{-\frac{1}{n}} dx$$

$$\ln x \frac{x^{\frac{1}{n}}}{\frac{1}{n}} = \int \cot(n\theta) d\theta$$

$$\ln x \frac{x^{\frac{1}{n}}}{\frac{1}{n}} = \frac{1}{n} \ln(\sin(n\theta)) + C$$

$$x^{\frac{1}{n}} = \frac{1}{n} \ln(\sin(n\theta)) + C$$

$$\Rightarrow x^n = \sin(n\theta) \cdot a^n$$

$$Ex) x^n \sin(n\theta) = a^n$$

$$x^{n-1} \sin(n\theta) \frac{dx}{d\theta} + x^n \cos(n\theta) (n) = 0$$

$$\frac{dx}{d\theta} = -x \cot(n\theta)$$

$$\frac{dx}{d\theta} \Rightarrow -x \frac{d\theta}{dx}$$

$$\Rightarrow \frac{x^{\frac{1}{n}}}{x^{\frac{1}{n}}} \frac{d\theta}{dx} = +x \cot(n\theta)$$

$$\frac{d\theta}{dx} = x^{\frac{1}{n}} \cot(n\theta) \times \frac{1}{x}$$

$$\int x^{-\frac{1}{n}} dx = \int \tan(n\theta) d\theta$$

$$\ln x \frac{x^{\frac{1}{n}}}{\frac{1}{n}} = \frac{1}{n} \ln(\sec(n\theta)) + C$$

$$x = (\sec(n\theta)) \cdot K$$

$$x^n = \sec(n\theta) \cdot K^n$$

$$x^n \cos(n\theta) = C$$

$$\frac{dy}{dx} = f(x, y), \quad y(0) = y_0$$

$$\text{or } \dot{y} = f(x, y), \quad y(0) = y_0$$

$$y = y_0 + \int_0^x f(x, y) dx$$

Ex) Solve the IVP $\dot{y} = y^2$, $y(0) = 1$,

$$f(t, y) = y^2$$

$$\text{Let } b=1, c=1$$

$$|f(t, y)| = y^2 \leq 4$$

$$\begin{aligned} |f(t, y_1) - f(t, y_2)| &= |y_1^2 - y_2^2| \\ &= |y_1 + y_2| |y_1 - y_2| \\ &\leq (|y_1| + |y_2|) |y_1 - y_2| \\ &\leq 4 |y_1 - y_2| \end{aligned}$$

$f(t, y)$ satisfies Lipschitz condition with $M=4$ and there exists a unique solution for $|t| \leq \min[1, \frac{1}{4}] = \frac{1}{4}$

The integral equation

$$y(t) = 1 + \int_0^t y^2 dt$$

Apply the method of successive approximation with $y=1$.

$$y_1(t) = 1 + \int_0^t 1 \cdot dt = 1+t$$

$$y_2(t) = 1 + \int_0^t (1+t)^2 dt = 1+t+t^2+\frac{t^3}{3}$$

$$y_3(t) = 1 + \int_0^t \left(1+t+t^2+\frac{t^3}{3}\right)^2 dt$$

$$= 1+t+t^2+t^3 + \frac{2}{3}t^4 + \frac{1}{3}t^5 + \frac{1}{9}t^6 + \frac{1}{63}t^7$$

Ex) Consider the IVP $\dot{y} = \sqrt{y}$, $y(0) = 0$

$$\frac{dy}{\sqrt{y}} = dt \quad \text{or} \quad 2\sqrt{y} = t + k$$

At $t=0$, $y=0$, $k=0$ $y = \frac{t^2}{4}$, $y = 0$

In this case, existence but not uniqueness of a solution.
 * Here $f(t, y) = \sqrt{y}$ does not satisfy a Lipschitz condition in
 $R = \{(t, y) \mid 0 \leq t \leq b, 0 \leq y \leq c\}$

Ex) $\alpha \pm i\beta$

$$C_1 e^{(\alpha + i\beta)x} + C_2 e^{(\alpha - i\beta)x} \quad A = C_1 + C_2$$

$$e^{\alpha x} (A \cos \beta x + B \sin \beta x) \quad B = i(C_1 - C_2)$$

If complex roots repeat,

Ex) $\frac{d^4 y}{dx^4} + 4 \frac{d^2 y}{dx^2} + 3y = 0$

$$y = e^{mx}, \quad \frac{d^k y}{dx^k} = m^k e^{mx}$$

$$m^4 + 4m^2 + 3 = 0$$

$$\text{or, } (m^2 + 3)(m^2 + 1) = 0$$

$$m = \pm \sqrt{3}i, \pm i$$

$$y = C_1 \cos \sqrt{3}x + C_2 \sin \sqrt{3}x + C_3 \cos x + C_4 \sin x$$

Ex) $\frac{d^4 y}{dx^4} + \frac{d^3 y}{dx^3} - 5 \frac{d^2 y}{dx^2} + 3 \frac{dy}{dx} = 0$

$$m^4 + m^3 - 5m^2 + 3m = 0$$

$$m(m^3 + m^2 - 5m + 3) = 0$$

$$m(m-1)^2(m+3) = 0$$

$$m = 0, 1, 1, -3$$

$$y = C_1 e^{0x} + C_2 e^{-3x} + (C_3 + C_4 x) e^x$$

$$y = C_1 + C_2 e^{-3x} + (C_3 + C_4 x) e^x$$

$$\begin{vmatrix} 1 & 1 & -5 & 3 \\ 0 & 1 & 2 & -3 \\ 1 & 2 & -3 & 0 \end{vmatrix}$$

$$0 \quad 1 \quad 2 \quad -3$$

$$1 \quad 2 \quad -3 \quad 0$$

$$m^2 + 2m - 3 = 0$$

$$m^2 + 3m - m - 3 = 0$$

$$m(m+3) - 1(m+3) = 0$$

Ex) Consider the differential equation

$$\frac{dy}{dx} + y = \tan x \quad \text{--- (1)}$$

$$\frac{dy}{dx} + y = 0 \quad \text{--- (2)}$$

$$y = e^{mx} \Rightarrow m + 1 = 0 \quad \text{(AE)}$$

$$m = -1$$

$$y = C_1 \cos x + C_2 \sin x$$

Assume

$$y_p(x) = v_1(x) \cos x + v_2(x) \sin x$$

is a solution of the differential equation (1) where the functions v_1 and v_2 are to be determined.

$$y_p'(x) = v_1(x)(-\sin x) + v_2(x) \cos x + v_1'(x) \cos x + v_2'(x) \sin x$$

We impose the condition.

$$v_1'(x) \cos x + v_2'(x) \sin x = 0 \quad \text{--- (3)}$$

$$y_p'(x) = v_1(x)(-\sin x) + v_2(x) \cos x$$

$$y_p''(x) = v_1'(x)(-\sin x) + v_2'(x) \cos x + v_1(x)(-\cos x) - v_2(x) \sin x$$

$$v_1'(x)(-\sin x) + v_2'(x) \cos x = \tan x \quad \text{--- (4)}$$

Solving, $[(3) \times \sin x] + [(4) \times \cos x]$

$$v_2'(x) \{ \sin^2 x + \cos^2 x \} = \tan x \cos x$$

$$\text{or } v_2'(x) = \sin x$$

$$v_2(x) = -\cos x$$

$$v_1'(x) = -v_2'(x) \frac{\sin x}{\cos x}$$

$$= -\frac{\sin^2 x}{\cos x}$$

$$v_1(x) = \frac{\cos^2 x - 1}{\cos x} = \cos x - \sec x$$

$$u_1(x) = \sin x - \log |\sec x + \tan x|$$

$$y_p = \sin x \cos x - \cos x \log |\sec x + \tan x| - \sin x \cos x$$

$$= -\cos x \log |\sec x + \tan x|$$

$$y'' + y = \tan x$$

$$y = y_c + y_p$$

$$= C_1 \cos x + C_2 \sin x$$

$$- \cos x \log |\sec x + \tan x|$$

Ex) Apply the method of variation of parameters to solve

$$\ddot{y} - y = e^{2x} \quad \text{--- ①}$$

Sol: The corresponding homogenous differential equation is

$$\ddot{y} - y = 0 \rightarrow \text{②}$$

Assume $y = e^{mt}$ is a solution of ② :

$$\text{A.E is } m^2 - 1 = 0$$

$$m = \pm 1$$

$$y_c = C_1 e^x + C_2 e^{-x}$$

Consider $y_p = v_1(x) e^x + v_2(x) e^{-x}$ as solution of ①

$$y_p' = v_1(x) e^x - v_2(x) e^{-x} + v_1'(x) e^x + v_2'(x) e^{-x}$$

$$v_1'(x) e^x + v_2'(x) e^{-x} = 0 \rightarrow \text{①}$$

$$y_p' = v_1(x) e^x - v_2(x) e^{-x}$$

$$y_p'' = v_1'(x) e^x - v_2'(x) e^{-x} + v_1(x) e^x + v_2(x) e^{-x}$$

Substitute y_p, y_p', y_p'' in the given differential equation ①

$$v_1'(x) e^x - v_2'(x) e^{-x} + v_1(x) e^x + v_2(x) e^{-x} - v_1(x) e^x - v_2(x) e^{-x} = e^{2x}$$

$$v_1'(x) e^x - v_2'(x) e^{-x} = e^{2x} \rightarrow \text{②}$$

Adding ① and ②,

$$2v_1'(x) e^x = e^{2x}$$

$$v_1'(x) = \frac{1}{2} \Rightarrow v_1(x) = \frac{x}{2}$$

$$v_2'(x) = -\frac{1}{2} e^{2x} \Rightarrow v_2(x) = -\frac{1}{4} e^{2x}$$

$$y_p = \frac{x}{2} e^x - \frac{1}{4} e^x$$

$$= \frac{(2x-1)}{4} e^x$$

$$y = y_c + y_p$$

$$= c_1 e^x + c_2 e^{-x} + \left(\frac{2x-1}{4}\right) e^x$$

Ex) Consider the differential equation

$$(x^2+1) \frac{d^2y}{dx^2} - 2x \frac{dy}{dx} + 2y = 6(x^2+1)^{3/2}$$

$$(x^2+1) \frac{d^2y}{dx^2} - 2x \frac{dy}{dx} + 2y = 0$$

$$y = x v(x) ; \quad \frac{dy}{dx} = v(x) + x v'(x)$$

$$\frac{d^2y}{dx^2} = 2v'(x) + x v''(x)$$

$$(x^2+1) \{ 2v'(x) + x v''(x) \} - 2x \{ x v'(x) + v(x) \} + 2x v(x) = 0$$

$$v'(x) \{ 2x^2 + 2 - 2x^2 \} + (x^2+1) x v''(x) = 0$$

$$2v'(x) + x(x^2+1)v''(x) = 0$$

$$p = v'(x)$$

$$2p + x(x^2+1) \frac{dp}{dx} = 0$$

$$\text{or, } \frac{dp}{p} + \frac{2dx}{x(x^2+1)} = 0$$

$$\text{or, } \frac{dp}{p} = \frac{-2dx}{x(x^2+1)}$$

$$\frac{dp}{p} = 2 \left\{ \frac{x}{(x^2+1)} - \frac{1}{x} \right\} dx \quad \log \frac{p}{c} = [\log(x^2+1) - \log x]$$

$$p = 1 + \frac{1}{x^2}$$

$$\log \frac{p}{c} = \log \frac{x^2+1}{x^2}$$

$$v'(x) = 1 + \frac{1}{x^2}$$

$$v(x) = x - \frac{1}{x}$$

$$y_2 = x^2 - 1$$

$$y_c = C_1 x + C_2 (x^2 - 1)$$

$$y_p = x^4 + 3x^2$$

$$y_p = x v_1(x) + (x^2 - 1) v_2(x)$$

$$x v_1'(x) + (x^2 - 1) v_2'(x) = 0 \rightarrow \textcircled{1}$$

$$v_1'(x) + 2x v_2'(x) = 6(x^2 + 1) \rightarrow \textcircled{2}$$

Solving ① and ②,

$$v_1'(x) = -6(x^2 - 1)$$

$$v_2'(x) = 6x$$

$$v_1(x) = -2x^3 + 6x$$

$$v_2(x) = 3x^2$$

Cauchy-Euler Equation:

$$x^n \frac{d^n y}{dx^n} + a_1 x^{n-1} \frac{d^{n-1} y}{dx^{n-1}} + a_2 x^{n-2} \frac{d^{n-2} y}{dx^{n-2}} + \dots + a_{n-1} x \frac{dy}{dx} + a_n y = f(x)$$

$$\text{Ex) } x^2 \frac{d^2 y}{dx^2} - 2x \frac{dy}{dx} + 2y = x^3 \rightarrow \textcircled{1}$$

$$\text{Let } x = e^t \Rightarrow t = \log x \Rightarrow \frac{dy}{dx} = \frac{dy}{dt} \cdot \frac{1}{x} \Rightarrow x \frac{dy}{dx} = \frac{dy}{dt}$$

$$x^2 \frac{d^2 y}{dx^2} = \frac{d}{dx} \left(\frac{dy}{dx} \right) = \frac{d}{dx} \left(\frac{dy}{dt} \cdot \frac{1}{x} \right)$$

$$\frac{d^2 y}{dx^2} = \frac{-1}{x^2} \frac{dy}{dt} + \frac{1}{x} \cdot \frac{d}{dx} \left(\frac{dy}{dt} \right)$$

$$= \frac{-1}{x^2} \frac{dy}{dt} + \frac{d}{dt} \left(\frac{dy}{dt} \right) \cdot \frac{dt}{dx} \cdot \frac{1}{x}$$

$$= \frac{-1}{x^2} \frac{dy}{dt} + \frac{d^2 y}{dt^2} \cdot \frac{1}{x^2}$$

$$x^2 \frac{d^2 y}{dx^2} = \frac{d^2 y}{dt^2} - \frac{dy}{dt}$$

$$\frac{d^2 y}{dt^2} - \frac{dy}{dt} - 2 \frac{dy}{dt} + 2y = e^{3t}$$

$$\frac{d^2 y}{dt^2} - 3 \frac{dy}{dt} + 2y = e^{3t} \rightarrow \textcircled{2}$$

Let $y = e^{mt}$ be a solution of ①

$$\frac{d^2y}{dt^2} - 3\frac{dy}{dt} + 2y = 0$$

Then the A.E is

$$m^2 - 3m + 2 = 0$$

$$(m-1)(m-2) = 0$$

$$m = 1, 2$$

$$y_c = C_1 e^t + C_2 e^{2t}$$

Let $y = v_1(t)e^t + v_2(t)e^{2t}$ be a ~~bo~~ solution of (2)

$$y_p' = v_1(t)e^t + v_2(t)(2e^{2t}) + v_1'(t)e^t + v_2'(t)e^{2t}$$

$$v_1'(t)e^t + v_2'(t)e^{2t} = 0 \longrightarrow \textcircled{I}$$

$$y_p' = v_1(t)e^t + v_2(t)(2e^{2t})$$

$$y_p'' = v_1'(t)e^t + 2v_2'(t)e^{2t} + v_1(t)e^t + 4v_2(t)e^{2t}$$

$$v_1'(t)e^t + 2v_2'(t)e^{2t} = e^{3t} \longrightarrow \textcircled{II}$$

Solving (I) and (II)

$$v_2'(t)e^{2t} = e^{3t}$$

y_p - particular integral.

$$v_2'(t) = e^t$$

$$v_2(t) = e^t$$

$$v_1'(t) = -e^{2t}$$

$$v_1(t) = -\frac{1}{2}e^{2t}$$

$$y_p = \frac{1}{2}e^{3t}$$

$$y = y_c + y_p = C_1 e^t + C_2 e^{2t} + \frac{1}{2}e^{3t}$$

$$= C_1 x + C_2 x^2 + \frac{1}{2}x^3$$

$$x \frac{dy}{dx} = \frac{dy}{dt} = Dy$$

$$D = \frac{d}{dt}$$

$$x^2 \frac{d^2y}{dx^2} = \frac{d^2y}{dt^2} - \frac{dy}{dt} = D^2y - Dy = D(D-1)y$$

$$x^3 \frac{d^3y}{dx^3} = \frac{d^3y}{dt^3} - 3\frac{d^2y}{dt^2} + 2\frac{dy}{dt} = D(D-1)(D-2)y$$

$$\text{Ex) } x^3 \frac{d^3 y}{dx^3} - 4x^2 \frac{d^2 y}{dx^2} + 8x \frac{dy}{dx} - 8y = 4 \log x$$

Assuming $x > 0$, let $x = e^t$, then $t = \log x$

$$\frac{dy}{dx} = \frac{1}{x} \frac{dy}{dt}$$

$$\frac{d^2 y}{dx^2} = \frac{1}{x^2} \left(\frac{d^2 y}{dt^2} - \frac{dy}{dt} \right)$$

$$\frac{d^3 y}{dx^3} = \frac{d}{dx} \left(\frac{d^2 y}{dx^2} \right) = \frac{1}{x^2} \frac{d}{dx} \left(\frac{d^2 y}{dt^2} - \frac{dy}{dt} \right) - \frac{2}{x^3} \left(\frac{d^2 y}{dt^2} - \frac{dy}{dt} \right)$$

$$= \frac{1}{x^2} \left(\frac{d^3 y}{dt^3} \cdot \frac{1}{x} - \frac{d^2 y}{dt^2} \cdot \frac{1}{x} \right) - \frac{2}{x^3} \left(\frac{d^2 y}{dt^2} - \frac{dy}{dt} \right)$$

$$= \frac{1}{x^3} \left(\frac{d^3 y}{dt^3} - \frac{d^2 y}{dt^2} \right) - \frac{2}{x^3} \left(\frac{d^2 y}{dt^2} - \frac{dy}{dt} \right)$$

$$x^3 \frac{d^3 y}{dx^3} = \frac{d^3 y}{dt^3} - 3 \frac{d^2 y}{dt^2} + 2 \frac{dy}{dt}$$

$$\frac{d^3 y}{dt^3} - 3 \frac{d^2 y}{dt^2} + 2 \frac{dy}{dt} - 4 \left(\frac{d^2 y}{dt^2} - \frac{dy}{dt} \right) + 8 \frac{dy}{dt} - 8y = 4t$$

$$\frac{d^3 y}{dt^3} - 7 \frac{d^2 y}{dt^2} + 14 \frac{dy}{dt} - 8y = 4t$$

$$\{ D(D-1)(D-2) - 4D(D-1) + 8D - 8 \} y = 4t$$

$$\{ D^3 - 3D^2 + 2D - 4D^2 + 4D + 8D - 8 \} y = 4t \quad \left[D = \frac{d}{dt} \right]$$

$$\{ D^3 - 7D^2 + 14D - 8 \} y = 4t$$

Consider $y = e^{mt}$ is a solution of $(D^3 - 7D^2 + 14D - 8)y = 0$

A.E is $m^3 - 7m^2 + 14m - 8 = 0$

$$(m-1)(m-2)(m-4) = 0$$

$$m = 1, 2, 4$$

$$y_c = C_1 e^t + C_2 e^{2t} + C_3 e^{4t}$$

$$3V_2'(t)e^{2t} + 9V_3'(t)e^{4t} = 0$$

$$3V_2'(t)e^{2t} + 15V_3'(t)e^{4t} = 4t$$

$$\left[t \frac{1}{4} e^{-4t} + \frac{1}{4} e^{-4t} \right] 6V_3'(t)e^{4t} = 4t$$

$$e^{-4t} \left[\frac{1}{4} e^{4t} \right] V_3'(t) = \frac{2t}{3} e^{-4t}$$

$$V_3(t) = \frac{e^{-4t}}{6} \left[t - \frac{1}{4} \right]$$

Solve

$$* a_n y^n + a_{n-1} y^{n-1} + \dots + a_1 \dot{y} + a_0 y = f(t)$$

$y_1(t), y_2(t), \dots, y_n(t)$ be a fundamental system of ~~equations~~
solutions of the associated homogenous equation.

~~$$y_p = A_1 y_1(t) + A_2 y_2(t) + \dots + A_n y_n(t)$$~~

$$y_p = A_1(t) y_1(t) + A_2(t) y_2(t) + \dots + A_n(t) y_n(t)$$

Differentiating, we have

$$\dot{y}_p = \dot{A}_1 y_1 + A_1 \dot{y}_1 + \dot{A}_2 y_2 + A_2 \dot{y}_2 + \dots + \dot{A}_n y_n + A_n \dot{y}_n$$

$$+ \dot{A}_1 y_1 + \dot{A}_2 y_2 + \dots + \dot{A}_n y_n$$

At this point, we put

$$\dot{A}_1 y_1 + \dot{A}_2 y_2 + \dots + \dot{A}_n y_n = 0 \rightarrow (1)$$

$$\ddot{y}_p = \ddot{A}_1 y_1 + A_1 \ddot{y}_1 + \ddot{A}_2 y_2 + A_2 \ddot{y}_2 + \dots + \ddot{A}_n y_n + A_n \ddot{y}_n$$

$$+ \dot{A}_1 \dot{y}_1 + \dot{A}_2 \dot{y}_2 + \dots + \dot{A}_n \dot{y}_n$$

$$\text{Assume, } \dot{A}_1 y_1 + \dot{A}_2 y_2 + \dots + \dot{A}_n y_n = 0 \rightarrow (2)$$

Continuing in this way,

$$y_p^{(n)} = A_1 y_1^{(n)} + A_2 y_2^{(n)} + \dots + A_n y_n^{(n)}$$

$$+ \dot{A}_1 y_1^{(n-1)} + \dot{A}_2 y_2^{(n-1)} + \dots + \dot{A}_n y_n^{(n-1)}$$

Substituting $y_p, \dot{y}_p, \dots, y_p^{(n)}$, we get

$$\dot{A}_1 y_1^{(n-1)} + \dot{A}_2 y_2^{(n-1)} + \dots + \dot{A}_n y_n^{(n-1)} = \frac{f(t)}{a_n(t)}$$

$$\dot{A}_1 y_1 + \dot{A}_2 y_2 + \dots + \dot{A}_n y_n = 0$$

$$\dot{A}_1 \dot{y}_1 + \dot{A}_2 \dot{y}_2 + \dots + \dot{A}_n \dot{y}_n = 0$$

$$\dot{A}_1 y_1^{(n-2)} + \dot{A}_2 y_2^{(n-2)} + \dots + \dot{A}_n y_n^{(n-2)} = 0$$

$$\dot{A}_1 y_1^{(n-1)} + \dot{A}_2 y_2^{(n-1)} + \dots + \dot{A}_n y_n^{(n-1)} = \frac{f(t)}{a_n(t)}$$

This is a system of n equations in n unknowns with the determinant of the coefficient matrix

$$W(t) = \begin{vmatrix} y_1 & y_2 & y_3 & \dots & y_n \\ \dot{y}_1 & \dot{y}_2 & \dot{y}_3 & \dots & \dot{y}_n \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ y_1^{(n-1)} & y_2^{(n-1)} & y_3^{(n-1)} & \dots & y_n^{(n-1)} \end{vmatrix}$$

which is the Wronskian of the fundamental system of solutions, $y_1, y_2, \dots, y_n$. The Wronskian is never zero.

Solve for $\dot{A}_1, \dot{A}_2, \dots, \dot{A}_n$, $k = 1, 2, \dots, n$.

$k = 1, 2, \dots, n$,

$$\dot{A}_k = \frac{f(t) W_k(t)}{a_n(t) W(t)}$$

where $W_k(t)$ is the determinant obtained from $W(t)$ by replacing the k^{th} column by $(0, 0, 0, \dots, 0, 1)$

A particular solution of the non-homogeneous equation is

$$y_p(t) = \sum_{k=1}^n A_k(t) y_k(t) = \sum_{k=1}^n y_k(t) \int_a^t \frac{f(t) W_k(t)}{a_n(t) W(t)} dt$$

Ex) Find the general solution of $y''' + 3y'' + 2y' = -e^{-t}$

Soln: $y''' + 3y'' + 2y' = -e^{-t}$

$$y''' + 3y'' + 2y' = 0$$

$$y = e^{mt}$$

AE is $m^3 + 3m^2 + 2m = 0$, or $m = 0, -1, -2$.

$$y_1 = 1, y_2 = e^{-t}, y_3 = e^{-2t}$$

$$W(t) = \begin{vmatrix} 1 & e^{-t} & e^{-2t} \\ 0 & -e^{-t} & -2e^{-2t} \\ 0 & e^{-t} & 4e^{-2t} \end{vmatrix} = -2e^{-3t}$$

$$W_1(t) = \begin{vmatrix} e^{-t} & e^{-2t} \\ -e^{-t} & -2e^{-2t} \\ e^{-t} & 4e^{-2t} \end{vmatrix} = -e^{-3t}$$

$$w_2(t) = \begin{vmatrix} 1 & 0 & e^{-2t} \\ 0 & 0 & -2e^{-2t} \\ 0 & 1 & 4e^{-2t} \end{vmatrix} = 2e^{-2t}$$

$$w_3(t) = \begin{vmatrix} 1 & e^{-t} & 0 \\ 0 & -e^{-t} & 0 \\ 0 & e^{-t} & 1 \end{vmatrix} = -e^{-t}$$

Therefore, $y_c = te^{-t} + \frac{1}{2}e^{-2t} - \frac{1}{2}e^{-t} = te^{-t}$

$$y = y_c + y_p = C_1 + C_2 e^{-t} + C_3 e^{-2t} + te^{-t}$$

Ex) $\frac{d^2 y}{dx^2} - y = e^{2x}$

$\frac{d^2 y}{dx^2} - y = 0$ | $AE \Rightarrow m^2 - 1 = 0$
 $m = \pm 1$

C.F. $y_c = C_1 e^x + C_2 e^{-x}$

Let $y = Ae^{2x}$ is a particular integral.

$$y_p' = 2Ae^{2x}$$

$$y_p'' = 4Ae^{2x}$$

$$4Ae^{2x} - Ae^{2x} = e^{2x}$$

$$3A = 1$$

$$A = 1/3$$

$$y_p = 1/3 e^{2x}$$

$$y = y_c + y_p$$

$$= C_1 e^x + C_2 e^{-x} + \frac{1}{3} e^{2x}$$

Ex) Solve by the method of undetermined coefficients

$$\frac{d^2 y}{dx^2} + b \frac{dy}{dx} + ay = 24e^{-3x} \rightarrow \textcircled{1}$$

Soln: Corresponding homogenous differential equation is

$$\frac{d^2y}{dx^2} + 6\frac{dy}{dx} + 9y = 0$$

The auxiliary equation is $m^2 + 6m + 9 = 0$

$$(m+3)^2 = 0$$

$$m = -3, -3$$

Complimentary Function (CF), $y_c = (C_1 + C_2x)e^{-3x}$

Let $y_p = Ax^2e^{-3x}$ be a particular integral of (1)

$$\frac{dy_p}{dx} = 2Ax e^{-3x} - 3Ax^2 e^{-3x} = (2-3x)Ax e^{-3x}$$

$$\frac{d^2y_p}{dx^2} = 2Ae^{-3x} - 6Ax e^{-3x} - 6Ax e^{-3x} + 9Ax^2 e^{-3x}$$

$$= (2 - 12x + 9x^2)Ae^{-3x}$$

$$Ae^{-3x} [2 - 12x + 9x^2 + 12x - 18x + 9x^2] = 24e^{-3x}$$

$$A[2] = 24 \Rightarrow A = 12$$

$$y_p = 12x^2e^{-3x}$$

$$y = y_c + y_p$$

$$= (C_1 + C_2x)e^{-3x} + 12x^2e^{-3x}$$

$$= (C_1 + C_2x + 12x^2)e^{-3x}$$

Method of Undetermined Coefficients:

$$\frac{d^2y}{dx^2} + P(x)\frac{dy}{dx} + Q(x)y = R(x)$$

$$y = C.F + y_p$$

To find a particular solution of the given equation

$$I. R(x) = e^{ax}$$

A. When a is not a root of A.E, take $y_p = Ae^{ax}$

B. When a is a simple root of A.E, take $y_p = Axe^{ax}$

C. When a is a double root of A.E, take $y_p = Ax^2e^{ax}$

I. When $R(x) = \cos ax$ or $\sin ax$

A. When $\sin ax$ or $\cos ax$ is not in C.F.,

$$y_p = A \sin ax + B \cos ax$$

B. When $\sin ax$ or $\cos ax$ is in C.F.,

$$y_p = x(A \sin ax + B \cos ax)$$

~~II. When $R(x) = \cos ax$ or $\sin ax$~~

III. When $R(x)$ is a polynomial or a constant, i.e.,

$$R(x) = a_0 + a_1 x + a_2 x^2 + \dots + a_n x^n$$

A. Take $y_p = A_0 + A_1 x + A_2 x^2 + \dots + A_n x^n$

B. When $Q(x) = 0$, take

$$y_p = x(A_0 + A_1 x + A_2 x^2 + \dots + A_n x^n)$$

IV. When $R(x) = e^{ax} \sin bx$ or

$$e^{ax} (a_0 + a_1 x + \dots + a_n x^n)$$

$$\sin bx (a_0 + a_1 x + \dots + a_n x^n) \text{ etc.,}$$

Modify y_p accordingly with the help of rules I, II, III.

Tutorial-II

i) Solve the following differential equations:

i) $(D^4 + 2D^3 - 3D^2 - 4D + 4)y = 0$

ii) $(D^4 - 81)y = 0$

iii) Using variation of parameters method, solve

$$\frac{d^2 y}{dx^2} + 2 \frac{dy}{dx} + y = \frac{e^{-x}}{x^2}$$

2) $x^2 \frac{d^2 y}{dx^2} - 2x \frac{dy}{dx} + 2y = x^3$ (Cauchy-Euler Equation)

3) $x^3 \frac{d^3 y}{dx^3} - 3x^2 \frac{d^2 y}{dx^2} + 6x \frac{dy}{dx} - 6y = 0$

Solutions

i) i) $m^4 + 2m^3 - 3m^2 - 4m + 4 = 0$

$$\begin{array}{cccccc} 1 & 2 & -3 & -4 & 4 & \\ 0 & 1 & 3 & 0 & -4 & \\ 1 & 3 & 0 & -4 & 0 & \end{array}$$

$$m^3 + 3m^2 - 4 = 0$$

$$\begin{array}{cccc} 1 & 3 & 0 & -4 \\ 0 & 1 & 4 & 4 \\ 1 & 4 & 4 & 0 \end{array}$$

$$m^2 + 4m + 4 = 0$$

$$(m+2)^2 = 0$$

$$m = 1, 1, -2, -2$$

$$y = y_c = (C_1 + C_2 x) e^x + (C_3 + C_4 x) e^{-2x}$$

$$\text{ii)} \quad m^4 = 81 \Rightarrow m = 3, -3, 3i, -3i$$

$$y = y_c = \cancel{(C_1 + C_2 x + C_3 x^2 + C_4 x^3) e^{3x}} \quad \left[C_1 e^{3x} + C_2 e^{-3x} + C_3 \cos 3x + C_4 \sin 3x \right]$$

$$\text{iii)} \quad m^2 + 2m + 1 = 0 \quad \left(\frac{d^2 y}{dx^2} + 2 \frac{dy}{dx} + y = 0 \right)$$

$$\frac{-2 \pm \sqrt{4-4}}{2} \Rightarrow m = -1, -1$$

$$y_c = (C_1 + C_2 x) e^{-x}$$

$$y_p = V_1(x) e^{-x} + V_2(x) x e^{-x}$$

$$y_p' = -V_1(x) e^{-x} + V_2(x) (e^{-x} - x e^{-x})$$

$$V_1'(x) e^{-x} + V_2'(x) x e^{-x} = 0 \rightarrow \textcircled{1}$$

$$y_p'' = V_1(x) e^{-x} - V_1'(x) e^{-x} + V_2'(x) (e^{-x} - x e^{-x}) - V_2(x) e^{-x} + V_2(x) (e^{-x} - x e^{-x})$$

$$= V_1(x) e^{-x} - 2V_2(x) e^{-x} + V_2(x) x e^{-x} - V_1'(x) e^{-x} + V_2'(x) e^{-x} - V_2'(x) x e^{-x}$$

Substituting,

$$\cancel{-V_1(x) e^{-x}} - \cancel{V_2(x) x e^{-x}} - V_1'(x) e^{-x} + V_2'(x) e^{-x} - V_2'(x) x e^{-x} = \frac{e^{-x}}{x^2}$$

$$\rightarrow \textcircled{2}$$

Solving $\textcircled{1}$ & $\textcircled{2}$

$$V_2'(x) (\cancel{x e^{-x}} + \cancel{e^{-x}} - \cancel{x e^{-x}}) = \frac{e^{-x}}{x^2}$$

$$V_2'(x) = \frac{1}{x^2}$$

$$V_2(x) = -\frac{1}{x}$$

$$V_1'(x) e^{-x} = -\frac{1}{x} x e^{-x} = -\frac{e^{-x}}{x}$$

$$V_1'(x) = -\frac{1}{x}$$

$$V_1(x) = -\ln x$$

$$y_p = -\ln x e^{-x} - \frac{1}{x} x e^{-x} = -e^{-x} (\ln x + 1)$$

$$y = y_c + y_p = (C_1 + C_2 x) e^{-x} - e^{-x} (\ln x + 1)$$

$$= e^{-x} (C_1 + C_2 x - \ln x - 1)$$

$$2) (D(D-1) - 2D + 2) y = x^3 e^{3x}$$

$$(D^2 - 3D + 2) y = x^3 e^{3x}$$

$$m^2 - 3m + 2 = 0$$

$$m = 1, 2$$

$$y_c = C_1 e^x + C_2 e^{2x}$$

$$y_p = V_1(x) e^x + V_2(x) e^{2x}$$

$$y_p' = V_1(x) e^x + 2V_2(x) e^{2x}$$

$$V_1'(x) e^x + V_2'(x) e^{2x} = 0 \rightarrow \textcircled{1}$$

$$V_1'(x) e^x + 2V_2'(x) e^{2x} = x^3 e^{3x} \rightarrow \textcircled{2}$$

Solving $\textcircled{1}$ and $\textcircled{2}$,

$$V_2'(x) (2e^{2x} - e^{2x}) = x^3 e^{3x}$$

$$V_2'(x) = x^3 e^{-2x} e^{3x} = e^{+x} \rightarrow V_2(x) = +e^{+x}$$

$$V_2(x) = \frac{-1}{2} x^3 e^{-2x} + \frac{1}{2} \int 3x^2 e^{-2x} dx$$

$$= \frac{-x^3}{2} e^{-2x} + \frac{3}{2} \left[\frac{-1}{2} x^2 e^{-2x} + \frac{1}{2} \int 2x e^{-2x} dx \right]$$

$$= \frac{-x^3}{2} e^{-2x} - \frac{3}{4} x^2 e^{-2x} + \frac{3}{2} \left[\frac{-1}{2} x e^{-2x} + \frac{1}{2} \int e^{-2x} dx \right]$$

$$= \frac{-1}{2} x^3 e^{-2x} - \frac{3}{4} x^2 e^{-2x} - \frac{3}{4} x e^{-2x} - \frac{3}{8} e^{-2x}$$

$$V_1'(x) = -V_2'(x) e^{3x} = -x^3 e^{-2x} e^{3x} = -x^3 e^{+x}$$

$$V_1(x) = \frac{-1}{4} x^4 - \frac{1}{2} e^{2x} = -\frac{e^{2x}}{2}$$

$$y_p = \frac{-1}{4} x^4 e^x + \frac{-1}{2} x^3 - \frac{3}{4} x^2 - \frac{3}{4} x - \frac{3}{8}$$

$$= \frac{-1}{4} (x^4 e^x + 2x^3 + 3x^2 + 3x - 3)$$

$$y_p = \frac{-e^{3x}}{2} + e^{3x}$$

$$y_p = \frac{1}{2} e^{3x}$$

$$y = y_c + y_p = C_1 e^x + C_2 e^{2x} - \frac{1}{8} (2x^4 e^x + 4x^3 + 6x^2 + 6x - 3)$$

$$y = C_1 e^x + C_2 e^{2x} + \frac{1}{2} e^{3x} = C_1 x + C_2 x^2 + \frac{1}{2} x^3$$

$$3) (D(D-1)(D-2) - 3D(D-1) + 6D - 6)y = 0$$

$$(D^3 - 3D^2 + 2D - 3D^2 + 3D + 6D - 6)y = 0$$

$$(D^3 - 6D^2 + 11D - 6)y = 0$$

$$m^3 - 6m^2 + 11m - 6 = 0$$

$$m = 1, 2, 3.$$

$$y = y_c = C_1 e^x + C_2 e^{2x} + C_3 e^{3x} = C_1 x + C_2 x^2 + C_3 x^3$$

4) $x \frac{d^2 y}{dx^2} - (x+1) \frac{dy}{dx} + y = x^2$ Given that $y_1 = e^x$, $y_2 = x+1$ are two solutions of the homogenous equation.

Soln: $\frac{d^2 y}{dx^2} - \frac{(x+1)}{x} \frac{dy}{dx} + \frac{1}{x} y = x$

$$y_p = v_1(x) e^x + v_2(x) (x+1)$$

$$y_p' = v_1(x) e^x + v_2(x)$$

$$v_1'(x) e^x + v_2'(x) (x+1) = 0 \rightarrow (1)$$

$$v_1'(x) e^x + v_2'(x) = x \rightarrow (2)$$

$$x + v_2'(x) \left(\cancel{x+1}^x \right) = 0$$

$$v_2'(x) = -x$$

$$v_2(x) = -\frac{x^2}{2}$$

$$v_1'(x) = \frac{x(x+1)}{e^x} = x(x+1)e^{-x}$$

$$v_1(x) = -(x+1)e^{-x} + \int e^{-x} = -e^{-x}(x+2)$$

$$y_p = -e^{-x}(x+2)e^x + (-x)(x+1)$$

$$= -x - 2 - x^2 - x$$

$$= -(x^2 + 2x + 2)$$

$$y = y_c + y_p = C_1 e^x + C_2(x+1) - x^2 - 2x - 2$$

$$= C_1 e^x + C_2(x+1) - x^2$$

$$5) \frac{d^2 y}{dx^2} - 3 \frac{dy}{dx} + 2y = \frac{e^x}{1+e^x}$$

$$m^2 - 3m + 2 = 0$$

$$m = 1, 2$$

$$y_c = c_1 e^x + c_2 e^{2x}$$

$$y_p = v_1(x) e^x + v_2(x) e^{2x}$$

$$y'_p = v_1(x) e^x + 2v_2(x) e^{2x}$$

$$v_1'(x) e^x + v_2'(x) e^{2x} = 0 \rightarrow \textcircled{1}$$

$$v_1'(x) e^x + 2v_2'(x) e^{2x} = \frac{e^x}{1+e^x}$$

$$v_2'(x) e^x = \frac{1}{1+e^x}$$

$$e^x = t, \quad e^x dx = dt$$

$$v_2(x) = \int \frac{1}{t(1+t)} \frac{dt}{t} \Rightarrow v_2(x) = \int \frac{1}{e^x(1+e^x)} dx$$

$$v_2(x) = \int \frac{1+e^x - e^x}{e^x(1+e^x)} dx = \int \left(\frac{1}{e^x} - \frac{e^x}{1+e^x} \right) dx$$

$$v_2(x) = -e^{-x} - \ln(1+e^x)$$

$$v_1'(x) = -v_2'(x) e^x = \frac{-e^{-x}}{1+e^x} e^x = \frac{-1}{1+e^x}$$

$$v_1(x) = \int \left(\frac{e^x}{1+e^x} - 1 \right) dx = \ln(1+e^x) - x$$

$$\begin{aligned} y_p &= (\ln(1+e^x) - x) e^x - (e^{-x} + \ln(1+e^x)) e^{2x} \\ &= (\ln(1+e^x) - x - 1) e^x - \ln(1+e^x) e^{2x} \end{aligned}$$

$$y = y_c + y_p = c_1 e^x + c_2 e^{2x} + e^x (\ln(1+e^x) - x) - \ln(1+e^x) e^{2x}$$

1) Prove that the system of confocal conics, $\frac{x^2}{a^2+\lambda} + \frac{y^2}{b^2+\lambda} = 1$, λ being the parameter is ~~differential~~ self-orthogonal.

Sol: ~~$(a^2+\lambda)2x \frac{dx}{d\lambda}$~~ $\frac{2x}{a^2+\lambda} + \frac{2y}{b^2+\lambda} \frac{dy}{d\lambda} = 0 \Rightarrow \left(\frac{dy}{dx} = -\frac{x}{a^2+\lambda} \times \frac{b^2+\lambda}{y} \right)$

Q.T $\Rightarrow \frac{dx}{dy} = \frac{a^2+\lambda}{b^2+\lambda} \times \frac{y}{x}$ $\ln x = \frac{b^2+\lambda}{a^2+\lambda} \ln y + C$ $a^2+\lambda = l$
 $b^2+\lambda = k$

$\ln \left(\frac{x}{b^2+\lambda} \times \frac{a^2+\lambda}{y} \right) = C$

$x^2 b^2 + \lambda(x^2 + y^2) + a^2 y^2 = (a^2+\lambda)(b^2+\lambda)$

$\frac{y}{x} \frac{dy}{dx} = -\frac{(b^2+\lambda)}{(a^2+\lambda)}$

$\lambda \left(1 + \frac{y}{x} \frac{dy}{dx} \right) = -\frac{(b^2 + \frac{y}{x} \frac{dy}{dx} a^2)}{1 + \frac{y}{x} \frac{dy}{dx}}$

$\frac{x}{a^2+\lambda} + \frac{y y'}{b^2+\lambda} = 0 \rightarrow (2)$

$\frac{x^2}{a^2+\lambda} + \frac{y^2}{b^2+\lambda} = 1 \rightarrow (1)$

$(1) - (2) \times x = \frac{y^2 - x y y'}{b^2+\lambda} = 1 \rightarrow \lambda = \frac{y^2 - x y y' - b^2}{y^2 - x y y'}$

$b^2 + \lambda = y^2 - x y y' \rightarrow (3)$

~~$\ln \left(\frac{x}{y} \times \frac{y^2 - x y y' + a^2 b^2}{a^2 + b^2} \right)$~~

$\frac{x^2}{a^2+\lambda} + \frac{y^2}{y^2 - x y y'} = 1 \Rightarrow \left(\frac{-x y y'}{y^2 - x y y'} \times \frac{1}{x^2} \right) = a^2 + \lambda$

$a^2 + \lambda = \frac{-x y + x y y'}{y^2} \rightarrow (4)$

$(3) - (4) \Rightarrow b^2 - a^2 = y^2 - x y y' + \frac{x y}{y^2} - \frac{x^2}{y^2}$

$b^2 - a^2 = y^2 - x^2 + x y \left(\frac{1}{y} - \frac{x}{y^2} \right)$

Replacing y' by $-\frac{dx}{dy}$, $y' \rightarrow -\frac{1}{y}$

$-\frac{dx}{dy} (b^2 - a^2) = -\frac{dx}{dy} (y^2 - x^2) + x y \left(1 - \left(\frac{dx}{dy} \right)^2 \right)$

$b^2 - a^2 = y^2 - x^2 + x y \left(-y' + \frac{1}{y} \right) \rightarrow (2)$

$(1) = (2)$ Hence, proved.

2) If S is defined by the rectangle $|x| \leq a, |y| \leq b$, show that function $f(x, y) = x \sin y + y \cos x$ satisfy Lipschitz condition. Find the Lipschitz constant.

Soln: $|f(x, y_1) - f(x, y_2)| = x \sin y_1 - x \sin y_2 + (y_1 - y_2) \cos x$
 $= 2x \cos\left(\frac{y_1 + y_2}{2}\right) \sin\left(\frac{y_1 - y_2}{2}\right) + (y_1 - y_2) \cos x$
 $\leq |y_1 - y_2| \left[x \cos\left(\frac{y_1 + y_2}{2}\right) + \cos x \right]$
 $\leq |y_1 - y_2| [a + 1]$
 Lipschitz constant $\rightarrow a + 1$

3) Can we drop the Lipschitz condition in the equation $f(x, y) = y^{2/3}$ on $R: |x| \leq 1, |y| \leq 1$.

Soln: $|f(x, y_1) - f(x, y_2)| = y_1^{2/3} - y_2^{2/3}$

Take $y_2 = 0$

$$|f(x, y_1) - f(x, y_2)| = |y_1^{2/3}| = \frac{|y_1 - 0|}{|y_1|^{1/3}}$$

$\frac{1}{|y_1|^{1/3}} \rightarrow \infty$ as $y_1 \rightarrow 0$ it is not bounded.

Can't drop the Lipschitz condition.

4) Find the largest interval in which Picard's theorem generates for unique solution $\frac{dy}{dx} = 16 + y^2; y(0) = 0$

Sol: $|x| < a, |y| < b \rightarrow$ Domain.

$f(x, y) = 16 + y^2 \rightarrow$ i) Since it is a polynomial, it is continuous.

ii) $|f(x, y)| = |16 + y^2| \leq 16 + |y|^2 \leq 16 + b^2$

Interval for the existence part

$$|x| < \min \left\{ a, \frac{b}{16 + b^2} \right\}$$

Uniqueness:

$$\frac{\partial f}{\partial y} = 2y, \quad \left| \frac{\partial f}{\partial y} \right| < 2b \Rightarrow f \text{ is Lipschitz}$$

$\Rightarrow$ solution is unique in $|x| < \min \left\{ a, \frac{b}{16 + b^2} \right\}$.

$$\text{Let } g(b) = \frac{b}{16+b^2}$$

$$g'(b) = \frac{16+b^2-2b^2}{(16+b^2)^2} = \frac{16-b^2}{(16+b^2)^2}$$

$$g'(b) = 0 \Rightarrow b = \pm 4 \Rightarrow \frac{b}{16+b^2} = \frac{1}{8}$$

If $a > \frac{1}{8}$

$|x| < \frac{1}{8} \rightarrow$ Region in which soln is unique

If $a < \frac{1}{8}$

$|x| < a \rightarrow$ "

5) Find the second approximation of the solution of the equation

$\frac{dy}{dx} = 2 - \frac{y}{x}$, $y(1) = 2$, by Picard's ~~theorem~~ method.

System of Linear Differential Equations:

$$\frac{dy}{dt} = 5x + 3y \quad \text{--- (i)}$$

$$\frac{dx}{dt} = 2x + y \quad \text{--- (ii)}$$

$$\frac{d^2y}{dt^2} = 5 \frac{dx}{dt} + 3 \frac{dy}{dt}$$

$$= 5(2x + y) + 3(5x + y) = 10x + 5y + 3 \frac{dy}{dt}$$

$$= 25x + 8y$$

$$\frac{d^2y}{dt^2} = 2 \left(\frac{dy}{dt} - 3y \right) + 5y + 3 \frac{dy}{dt}$$

$$\frac{d^2y}{dt^2} = 5 \frac{dy}{dt} - y$$

$$\frac{d^2y}{dt^2} - 5 \frac{dy}{dt} + y = 0$$

$$y = e^{mt}$$

$$\text{A.E is } m^2 - 5m + 1 = 0$$

$$m = \frac{5 \pm \sqrt{21}}{2}$$

$$y = C_1 e^{\left(\frac{5+\sqrt{21}}{2}\right)t} + C_2 e^{\left(\frac{5-\sqrt{21}}{2}\right)t}$$

$$\text{From (i) } 5x = \frac{dy}{dt} - 3y$$

$$x = \frac{1}{5} \left[\frac{dy}{dt} - 3y \right]$$

$$\text{Ex) } (5D+4)x - (2D+1)y = e^{-t} \quad \text{--- (i) } D = \frac{d}{dt}$$

$$(D+8)x - 3y = 5e^{-t} \quad \text{--- (ii)}$$

Soln: Multiply eqn ① by ③ and operate eqn ② by (2D+1)

$$3((5D+4)x) - 3(2D+1)y = 3e^{-t}$$

$$(2D+1)(D+8)x - 3(2D+1)y = 5(2D+1)e^{-t}$$

$$\{3(5D+4) - (2D+1)(D+8)\}x = (3 - 10D - 5)e^{-t}$$

$$(-2D^2 - 2D + 4)x = 8e^{-t}$$

$$-8e^{-t} + 10e^{-t}$$

$$(D^2 + D - 2)x = -4e^{-t}$$

$$\frac{d^2x}{dt^2} + \frac{dx}{dt} - 2x = -4e^{-t} \rightarrow (iii)$$

Let $x = e^{mt}$ is a solution of

$$\frac{d^2x}{dt^2} + \frac{dx}{dt} - 2x = 0$$

A.E is $m^2 + m - 2 = 0$

$$m = \frac{-1 \pm 3}{2} = -2, 1$$

C.F $x_c = C_1 e^{-2t} + C_2 e^t$

$x_p = A e^{-t}$ be a particular soln of (iii)

$$\frac{dx_p}{dt} = x_p' = -A e^{-t}, \quad \frac{d^2x_p}{dt^2} = A e^{-t}$$

$$A e^{-t} - A e^{-t} - 2A e^{-t} = -4e^{-t}$$

$$A e^{-t} = 2e^{-t}$$

$$A = 2$$

$$x_p = 2e^{-t}$$

$$x = x_c + x_p$$

$$= C_1 e^{-2t} + C_2 e^t + 2e^{-t}$$

From (ii),

$$(D+8)\{C_1 e^{-2t} + C_2 e^t + 2e^{-t}\} - 3y = 5e^{-t}$$

$$\Rightarrow -2e^{-t} + C_2 e^t - 2C_1 e^{-2t} + 8C_1 e^{-2t} + 8C_2 e^t + 16e^{-t} - 3y = 5e^{-t}$$

$$3y = 9e^{-t} + 9C_2 e^t + 6C_1 e^{-2t}$$

$$y = 3e^{-t} + 3C_2 e^t + 2C_1 e^{-2t}$$

Consider the n^{th} order linear differential equation

$$y^{(n)} + a_1 y^{(n-1)} + a_2 y^{(n-2)} + \dots + a_{n-1} \dot{y} + a_n y = f(t)$$

If we introduce the variables,

$$u_1 = y, u_2 = \dot{y}, u_3 = \ddot{y}, \dots, u_n = y^{(n-1)}$$

then the single n^{th} order equation is equivalent to the first order system

$$\dot{u}_1 = u_2, \dot{u}_2 = u_3, \dots, \dot{u}_{n-1} = u_n$$

$$\dot{u}_n = f(t) - a_n u_1 - a_{n-1} u_2 - \dots - a_1 u_n$$

* Study of first order system of the form

$$\dot{y}_1(t) = f_1(t, y_1, y_2, \dots, y_n)$$

$$\dot{y}_2(t) = f_2(t, y_1, y_2, \dots, y_n)$$

$$\vdots$$

$$\dot{y}_n(t) = f_n(t, y_1, y_2, \dots, y_n)$$

Tutorial

1) Find the curve $y(x)$ through the origin for which $y'' = y'$ and the tangent at the origin is $y = x$.

Soln:

$$\frac{d^2 y}{dx^2} = \frac{dy}{dx} \Rightarrow \frac{d^2 y}{dx^2} - \frac{dy}{dx} = 0$$

$$m^2 - m = 0 \Rightarrow m = 0, 1$$

$$y = c_1 e^t + c_2$$

$$y'(0) = 1$$

$$c_1 e^t \Big|_{t=0} = 1 \Rightarrow c_1 = 1$$

$$y(0) = 0$$

$$\Rightarrow c_1 + c_2 = 0$$

$$c_2 = -1$$

$$\boxed{y = e^x - 1}$$

2) Are the following pairs of functions linearly independent on the given interval?

i) $x^3, x^2|x|$; $-1 < x < 1$

ii) $x|x|, x^2$; $0 \leq x \leq 1$

Soln: i) for $-1 < x < 1$ $x^3 = x^3$

$$\text{for } \begin{cases} -1 < x < 0 : x^2|x| = -x^3 \\ 0 \leq x < 1 : x^2|x| = x^3 \end{cases}$$

$$-1 < x < 0 : c_1 x^3 + c_2 (-x^3) = 0 \Rightarrow x^3 (c_1 - c_2) = 0 \Rightarrow c_1 = c_2$$

$$0 \leq x < 1 : c_1 x^3 + c_2 (x^3) = 0 \Rightarrow x^3 (c_1 + c_2) = 0 \Rightarrow c_1 = -c_2$$

functions are linearly dependent.

ii) $x|x| = x^2$ for $0 \leq x \leq 1 \Rightarrow$ linearly dependent.

3) Solve the initial value problem:

$$(x^2 D^2 - 5x D + 8)y = 0, \quad y(1) = 5, \quad y'(1) = 18.$$

$$x = e^t$$

$$(D(D-1) - 5D + 8)y = 0$$

$$(D^2 - 6D + 8)y = 0$$

$$D = \frac{d}{dt}$$

$$m^2 - 6m + 8 = 0$$

$$m = \frac{6 \pm 2}{2} = 4, 2$$

$$y(1) = C_1 + C_2 = 5$$

$$y'(1) = 4C_1 + 2C_2 = 18$$

$$y = C_1 e^{4t} + C_2 e^{2t}$$

$$y = C_1 x^4 + C_2 x^2$$

$$\begin{cases} 2C_1 = 8 \\ C_1 = 4 \end{cases} \quad C_2 = 1$$

$$\boxed{y = 4x^4 + x^2}$$

4) Using the method of undetermined coefficients, determine the particular solution. Also find the general solution of

$$y^{(iv)} + y = 6 \sin x$$

$$m^4 + 1 = 0 \Rightarrow m = \pm i \Rightarrow m = \pm i^{1/2}, \pm i^{3/2}$$

$$\text{Let } y = A \sin x$$

$$y' = A \cos x \mid y'' = -A \sin x \mid y''' = -A \cos x \mid y^{(iv)} = A \sin x$$

$$2A \sin x = 6 \sin x$$

$$A = 3$$

$$\boxed{y = 3 \sin x}$$

$$w(t) = \begin{vmatrix} e^{\sqrt{x}} & e^{-\sqrt{x}} & e^{i^{3/2}x} & e^{-i^{3/2}x} \\ \sqrt{x} e^{\sqrt{x}} & -\sqrt{x} e^{-\sqrt{x}} & i^{3/2} e^{i^{3/2}x} & -i^{3/2} e^{-i^{3/2}x} \\ i e^{\sqrt{x}} & -i e^{-\sqrt{x}} & -e^{i^{3/2}x} & e^{-i^{3/2}x} \\ \frac{3}{2} \sqrt{x} e^{\sqrt{x}} & -\frac{3}{2} \sqrt{x} e^{-\sqrt{x}} & i^{3/2} e^{i^{3/2}x} & -i^{3/2} e^{-i^{3/2}x} \end{vmatrix}$$

$$\begin{vmatrix} 1 & 0 & 1 & 0 \\ \sqrt{x} & -\sqrt{x} & i^{3/2} & -i^{3/2} \\ i & -i & -1 & 1 \\ \frac{3}{2} \sqrt{x} & -\frac{3}{2} \sqrt{x} & i^{3/2} & -i^{3/2} \end{vmatrix}$$

5) Solve the Cauchy-Euler equation

$$x^2 y'' + 2xy' - 6y = 0$$

Soln:

$$(D(D-1) + 2D - 6)y = 0$$

$$(D^2 + D - 6)y = 0$$

$$m = \frac{-1 \pm 5}{2} = -3, 2$$

$$y = C_1 x^{-3} + C_2 x^2$$

6) Using the method of variation of parameters, determine a particular solution for

$$y'' + 4y' + 4y = x^{-2} e^{-2x}, \quad x > 0$$

$$0 = 4 - m^2 - 4m \quad \text{if } y = e^{mx}$$

$$1 - m = \frac{2 \pm 5}{2} = m$$

$$\begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix} = 0$$

$$\begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix} \frac{1}{3} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I$$

$$\begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix} \frac{1}{3} = I$$

$$\begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix} = 3I$$

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} A + \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} B = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$A \begin{pmatrix} 1 \\ 1 \end{pmatrix} = Y \quad B \begin{pmatrix} 1 \\ 1 \end{pmatrix} = Y$$

$$\begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix} = A$$

$$0 = 1 - 1 - 1 = -1 \quad \text{if } y = e^{mx}$$

$$0 = 1 - 1 - 1 = -1 \quad \text{if } y = e^{mx}$$

$$0 = 1 - 1 - 1 = -1$$

$$1 - 1 = 0$$

Ex) Find the general solution of the system

$$\dot{y}_1 = y_1 + 2y_2$$

$$\dot{y}_2 = 3y_1 + 2y_2$$

$$y = \begin{pmatrix} y_1 \\ y_2 \end{pmatrix}$$

$$\dot{y} = \begin{pmatrix} \dot{y}_1 \\ \dot{y}_2 \end{pmatrix}$$

$$\dot{y} = Ay$$

Soln: Eliminate y_2 by the following way

$$\dot{y}_2 \neq \text{Bth} \quad \ddot{y}_1 = \dot{y}_1 + 2\dot{y}_2 \\ = \dot{y}_1 + 2(3y_1 + 2y_2)$$

$$\ddot{y}_1 = \dot{y}_1 + 6y_1 + 4y_2$$

$$\ddot{y}_1 = \dot{y}_1 + 6y_1 + 2(\dot{y}_1 - y_1)$$

$$\ddot{y}_1 = 3\dot{y}_1 + 4y_1$$

$$\ddot{y}_1 - 3\dot{y}_1 - 4y_1 = 0 \rightarrow \textcircled{1}$$

Let $y_1 = e^{mt}$ be a solution of $\textcircled{1}$.

The A.E is $m^2 - 3m - 4 = 0$

$$m = \frac{3 \pm 5}{2} = 4, -1$$

$$y_1 = c_1 e^{4t} + c_2 e^{-t}$$

$$y_2 = \frac{1}{2} [\dot{y}_1 - y_1] = \frac{1}{2} [4c_1 e^{4t} - c_2 e^{-t} - c_1 e^{4t} - c_2 e^{-t}]$$

$$y_2 = \frac{1}{2} [3c_1 e^{4t} - 2c_2 e^{-t}]$$

$$y_2 = \frac{3}{2} c_1 e^{4t} - c_2 e^{-t}$$

$$\begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = c_2 \begin{pmatrix} e^{-t} \\ -e^{-t} \end{pmatrix} + c_1 \begin{pmatrix} e^{4t} \\ \frac{3}{2} e^{4t} \end{pmatrix}$$

Consider the vectors,

$$y_1 = \begin{pmatrix} 1 \\ \frac{3}{2} \end{pmatrix} e^{4t} ; y_2 = \begin{pmatrix} 1 \\ -1 \end{pmatrix} e^{-t}$$

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 2 \end{pmatrix}$$

The characteristic equation is $|A - \lambda I| = 0$

$$\begin{vmatrix} 1-\lambda & 2 \\ 3 & 2-\lambda \end{vmatrix} = 0 \Rightarrow \lambda^2 - 3\lambda + 2 - 6 = 0$$

$$\lambda^2 - 3\lambda - 4 = 0$$

$$\lambda = 4, -1$$

The vectors $\begin{pmatrix} 1 \\ 3/2 \end{pmatrix}$ and $\begin{pmatrix} 1 \\ -1 \end{pmatrix}$ are the corresponding characteristic vectors.

Ex) Find the general solution of the system

$$\dot{y}_1 = 9y_1 - 3y_2$$

$$\dot{y}_2 = -3y_1 + 12y_2 - 3y_3$$

$$\dot{y}_3 = -3y_2 + 9y_3$$

$$Y = \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix}, \quad \dot{Y} = \begin{pmatrix} \dot{y}_1 \\ \dot{y}_2 \\ \dot{y}_3 \end{pmatrix}$$

$$\dot{Y} = MY$$

Soln. We look for solution of the form

$$Y = X e^{\lambda t}$$

$$\lambda X e^{\lambda t} = M X e^{\lambda t}$$

$$\dot{Y} = AY + B$$

Substituting, we get

$$(\lambda X - MX) e^{\lambda t} = 0$$

$$\text{or, } \lambda X = MX$$

$$(\lambda I - M)X = 0$$

For the non-trivial solution of ②, we have

$$|M - \lambda I| = 0$$

$$\begin{vmatrix} 9-\lambda & -3 & 0 \\ -3 & 12-\lambda & -3 \\ 0 & -3 & 9-\lambda \end{vmatrix} = 0$$

$$(9-\lambda)(\lambda^2 - 21\lambda + 108 - 9) - 3(3)(9-\lambda) = 0$$

$$\Rightarrow (9-\lambda)(\lambda^2 - 21\lambda + 90) = 0$$

$$\Rightarrow (9-\lambda)(\lambda-6)(\lambda-15) = 0$$

$$\lambda = 6, 9, 15$$

$$\lambda_1 = 6, \lambda_2 = 9, \lambda_3 = 15$$

The corresponding characteristic vectors are

$$X_1 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, X_2 = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}, X_3 = \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix}$$

The general soln is therefore,

$$Y = C_1 \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} e^{6t} + C_2 \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} e^{9t} + C_3 \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix} e^{15t}$$

* Method for finding particular integrals in special cases:

Consider $F(D)y = f(x)$ is differential equation with constant coefficients $(D = \frac{d}{dx})$

$$y = \frac{1}{F(D)} f(x)$$

I) $f(x) = e^{mx}$

Then $y = \frac{1}{F(D)} f(x) = \frac{1}{F(D)} e^{mx} = \frac{1}{F(m)} e^{mx}$ if $F(m) \neq 0$

Ex) $(D^2 - 1)y = e^{2x}$

$$y = \frac{1}{D^2 - 1} e^{2x} = \frac{1}{2^2 - 1} e^{2x} = \frac{1}{3} e^{2x}$$

$$\frac{d^2 y}{dx^2} + 4y = (D^2 + 4)y$$

$$= \sin ax / e^{ax} / 2x e^x$$

II) $f(x) = e^{mx} v$

$$y = \frac{1}{F(D)} e^{mx} v$$

$$= e^{mx} \frac{1}{F(D+m)} v$$

III) $f(x) = \sin ax$ or $\cos ax$, $F(D) = \phi(a^2)$, $F(D)$ is an even polynomial in D

$$y = \frac{1}{F(D)} \sin ax$$

$$= \frac{1}{\phi(-a^2)} \sin ax \quad (\text{if } \phi(-a^2) \neq 0)$$

or, $y = \frac{1}{F(D)} \cos ax = \frac{1}{\phi(-a^2)} \cos ax \quad (\text{if } \phi(-a^2) \neq 0)$

$$D^2(\sin ax) = -a^2 \sin ax$$

$$D^4(\sin ax) = (-a^2)^2 \sin ax$$

$$D^6(\sin ax) = (-a^2)^3 \sin ax$$

Since $\phi(D^2)$ is a polynomial in D^2

$$\phi(D) \sin ax = \phi(-a^2) \sin ax$$

$$\sin ax = \frac{1}{F(D)} \phi(-a^2) \sin ax$$

$$= \phi(a^2) \frac{1}{F(D)} \sin ax$$

$$\text{or, } \frac{1}{F(D)} \sin ax = \frac{1}{\phi(a^2)} \sin ax$$

Ex) Solve $(D^2 - 4)y = \sin 2x$

Sol: $F(D) = D^2 - 4 = (D+2)(D-2)$

$$(D^2 - 4)y = 0$$

AE: $m^2 - 4 = 0 \Rightarrow m = \pm 2$

CF: $y = C_1 e^{2x} + C_2 e^{-2x}$

$$y_p = \frac{1}{D^2 - 4} \sin 2x$$

$$= \frac{1}{-2^2 - 4} \sin 2x = -\frac{1}{8} \sin 2x$$

$$y = y_c + y_p$$

$$= C_1 e^{2x} + C_2 e^{-2x} - \frac{1}{8} \sin 2x$$

Ex) Solve $(3D^2 + 2D - 8)y = 5 \cos x$

Sol: The associated homogenous differential eqn is

$$(3D^2 + 2D - 8)y = 0$$

AE is $3m^2 + 2m - 8 = 0$

$$m = \frac{-2 \pm 10}{6} = \frac{4}{3}, -2$$

CF $\rightarrow y_c = C_1 e^{\frac{4}{3}x} + C_2 e^{-2x}$

PI $\rightarrow y_p = \frac{1}{3D^2 + 2D - 8} 5 \cos x = \frac{1}{(D+2)(3D-4)} 5 \cos x$

$$= \frac{5(3D+4)(D-2)}{(D^2-4)(9D^2-16)} \times \cos x$$

$$= \frac{5(3D+4)(D-2)}{(-1-4)(-9-16)} \times \cos x = \frac{1}{25} (3D+4)(D-2) \cos x$$

$$= \frac{1}{25} (3D^2 - 2D - 8) \cos x$$

$$= \frac{1}{25} (-3\cos x + 2\sin x - 8\cos x)$$

$$= \frac{1}{25} (2\sin x - 11\cos x)$$

$$y = C_1 e^{-2x} + C_2 e^{1/3 x} + \frac{1}{25} (2\sin x - 11\cos x)$$

IV) Suppose $f(x) = x^m$, m is a +ve integer

$$\frac{1}{F(D)} x^m = (a_0 + a_1 D + a_2 D^2 + \dots + a_m D^m) x^m$$

Ex) Solve $(D^4 - 2D^3 + D^2)y = x^3$

Soln: AE $\rightarrow m^4 - 2m^3 + m^2 = 0$

$$m = 1, 1, 0, 0$$

CF $\rightarrow y_c = (C_1 + C_2 x)e^x + (C_3 + C_4 x)$

PI $\rightarrow y_p = \frac{1}{F(D)} x^m$

$$= \frac{1}{D^4 - 2D^3 + D^2} (x^3) = \frac{1}{D^2(D-1)^2} (x^3)$$

$$= \frac{1}{D^2} (1-D)^{-2} x^3 = \frac{1}{D^2} (1+2D) x^3$$

$$= \frac{1}{D^2} (1 + 2D + 3D^2 + 4D^3) x^3$$

$$= \frac{1}{D^2} (x^3 + 6x^2 + 18x + 24)$$

$$= \frac{1}{D} \left(\frac{x^4}{4} + 2x^3 + 9x^2 + 24x \right)$$

$$= \frac{x^5}{20} + \frac{x^4}{2} + 3x^3 + 12x^2$$

$$y = y_c + y_p = (C_1 + C_2 x) + (C_3 + C_4 x) e^x + \frac{x^5}{20} + \frac{x^4}{2} + 3x^3 + 12x^2$$

For v , suppose $f(x) = xv$ or $x^m v$

i)